

14. Which one of the following factors affects the electrical resistivity of a conductor ?

- (A) Temperature
- (B) Length
- (C) Shape
- (D) Thickness

CBSE 2025 C

15. Four wires a, b, c and d are made of the same materials. If ' l ' denotes length and ' r ' denotes radius, then the wire which offers least resistance to the flow of electric current has length and radius respectively as :

(A) $2l$ and $4r$

(B) l and $2r$

(C) $4l$ and $\frac{r}{4}$

(D) $\frac{l}{4}$ and $\frac{r}{2}$

CBSE 2025 C

- 25.** State Ohm's law. Give reason why the shape of voltage versus current ($V - I$) graph for a conductor should be a straight line passing through the origin.

CBSE 2025 C

- (b) (i) Define the term electric power. Express it in terms of potential difference (V) and resistance (R).
- (ii) The resistance of a metal wire of length 6 m is $120\ \Omega$. If the area of cross-section of the wire is $4 \times 10^{-7}\ \text{m}^2$, calculate the electrical resistivity of the material of the wire. How would the resistivity be affected if this wire is gradually stretched so that its length is doubled ? Justify your answer.

CBSE 2025 C

19. *Assertion (A) :* The bending of a wire in the form of a coil of several turns does not affect its electrical resistance.

Reason (R) : The electrical resistance of a wire is directly proportional to its electrical resistivity.

CBSE 2025

14. The resistivity of a wire made of an alloy is generally :
- (A) Lower than that of its constituent metals.
 - (B) Higher than that of its constituent metals.
 - (C) Decreases with increase in its area of cross-section.
 - (D) Increases with increase in its length.

CBSE 2025

14. The resistivity of a wire made of an alloy is generally :
- (A) Lower than that of its constituent metals.
 - (B) Higher than that of its constituent metals.
 - (C) Decreases with increase in its area of cross-section.
 - (D) Increases with increase in its length.

CBSE 2025

(B) / Higher than that of its constituent metals.

18. *Assertion (A) :* Nichrome is an alloy which is commonly used in electrical heating devices such as electric irons, toasters, etc.

Reason (R) : The resistivity of nichrome is high and its resistance decreases with increase in temperature.

CBSE 2025

18. *Assertion (A) :* Nichrome is an alloy which is commonly used in electrical heating devices such as electric irons, toasters, etc.

Reason (R) : The resistivity of nichrome is high and its resistance decreases with increase in temperature.

CBSE 2025

(C) /A is true, but R is false.

- 32.** Resistance of a wire of length 1 m is $35\ \Omega$ at 20°C . If the diameter of the wire is 0.2 mm, determine the resistivity of the material of the wire at that temperature. How will the resistivity of the wire change if the length and diameter of the wire both are doubled ? Justify your answer.

(Given that $\pi = \frac{22}{7}$)

CBSE 2025

3

$$R = 35 \, \Omega$$

$$\text{Diameter } (2r) = 0.2 \, \text{mm} = 2 \times 10^{-4} \, \text{m} \quad \therefore r = 10^{-4} \, \text{m}$$

$$\text{Length of the wire} = 1 \, \text{m}$$

$$R = \rho \frac{l}{A}$$

$$\text{Area of cross section, } A = \pi r^2$$

$$\text{Resistivity, } \rho = \frac{RA}{l} = \frac{35 \times \frac{22}{7} \times (10^{-4})^2}{1}$$

$$= 110 \times 10^{-8} \, \Omega\text{m}$$

Resistivity of the wire will not change

as it is a characteristic property of the material which does not depend on the dimensions of the wire.

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

33. (a) Write the relationship between resistivity and resistance of a cylindrical conductor of length l and area of cross-section A . Hence derive the SI unit of resistivity.

(b) Why are alloys used in electrical heating devices ?

CBSE 2025

33. (a) Write the relationship between resistivity and resistance of a cylindrical conductor of length l and area of cross-section A . Hence derive the SI unit of resistivity.

(b) Why are alloys used in electrical heating devices ?

CBSE 2025

3

(a)

- $R = \rho \frac{l}{A}$

$$\rho = \frac{RA}{l}$$

- SI unit of $\rho = \frac{\text{ohm} \times (\text{metre})^2}{\text{metre}}$
= ohm meter

1

$\frac{1}{2}$

$\frac{1}{2}$

(b) Resistivity of alloy is higher than pure metals. / They do not oxidise (burn) readily at high temperatures.

1

- 27.** (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is $14\ \Omega$. If the resistivity of the material of the wire is $44 \times 10^{-8}\ \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)

CBSE 2025

27. (a) Define one ampere.
- (b) The resistance of a wire of 0.01 cm radius is $14 \, \Omega$. If the resistivity of the material of the wire is $44 \times 10^{-8} \, \Omega\text{m}$, find the length of the wire. (Given $\pi = \frac{22}{7}$)

CBSE 2025

3

(a) When 1 coulomb of charge flows through any cross section of a conductor in 1 second, then the electric current flowing through it is 1 ampere. / $1\text{A} = \frac{1\text{C}}{1\text{s}}$

• (b) Radius of wire = $0.01 \, \text{cm} = 0.01 \times 10^{-2} \, \text{m}$

• $\rho = 44 \times 10^{-8} \, \Omega\text{m}$

• $R = 14 \, \Omega$

• $R = \rho \frac{l}{A}$

• $1 = \frac{R \times A}{\rho} = \frac{14 \times 22 \times (0.01 \times 10^{-2})^2}{7 \times 44 \times 10^{-8}}$

• $= \frac{10^{-8}}{10^{-8}}$
 $= 1.0 \, \text{m}$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

32. The electrical resistivity of three materials A, B and C at 20°C is given below :

CBSE 2025

3

Material	Resistivity ($\Omega \text{ m}$)
A	10^{17}
B	44×10^{-6}
C	1.62×10^{-8}

- Classify these materials as conductor, alloy and insulator.
- Give one example of each of these materials and state one use of each material in the design of an electrical appliance say an electric stove or an electric iron.

(i)

A - Insulator

$\frac{1}{2}$

B - Alloy

$\frac{1}{2}$

C - Conductor

$\frac{1}{2}$

(ii)

A: Plastic - handle of an electric iron.

B: Nichrome – used as a heating element in an electric iron.

C: Copper - electric wires. /

A: Rubber– foot of the electric stove.

B: Nichrome – used as a heating element in an electric stove.

C: Copper- electric wires.

(any other example with its use in an electric appliance)

($\frac{1}{2}$ X3)

11. A wire of length ' l ' is gradually stretched so that its length increases to $3l$. If its original resistance is R , then its new resistance will be :

CBSE 2025

(A) $3R$

(B) $6R$

(C) $9R$

(D) $27R$

11. A wire of length ' l ' is gradually stretched so that its length increases to $3l$. If its original resistance is R , then its new resistance will be : **CBSE 2025**

(A) $3R$

(B) $6R$

(C) $9R$

(D) $27R$

(C) / $9R$

32. Define the term “potential difference” between two points in an electric circuit carrying current. Name and define its S.I. unit. Also express it in terms of S.I. unit of work and charge.

CBSE 2025

32. Define the term “potential difference” between two points in an electric circuit carrying current. Name and define its S.I. unit. Also express it in terms of S.I. unit of work and charge.

CBSE 2025

3

- The work done to move a unit charge from one point to other in a conductor. / ($V=W/Q$)
- volt (V)
- In a current carrying conductor, when one joule of work is done to move a charge of 1 coulomb from one point to another.
- 1 volt = 1 joule/1 coulomb or $1V=1 \text{ J C}^{-1}$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

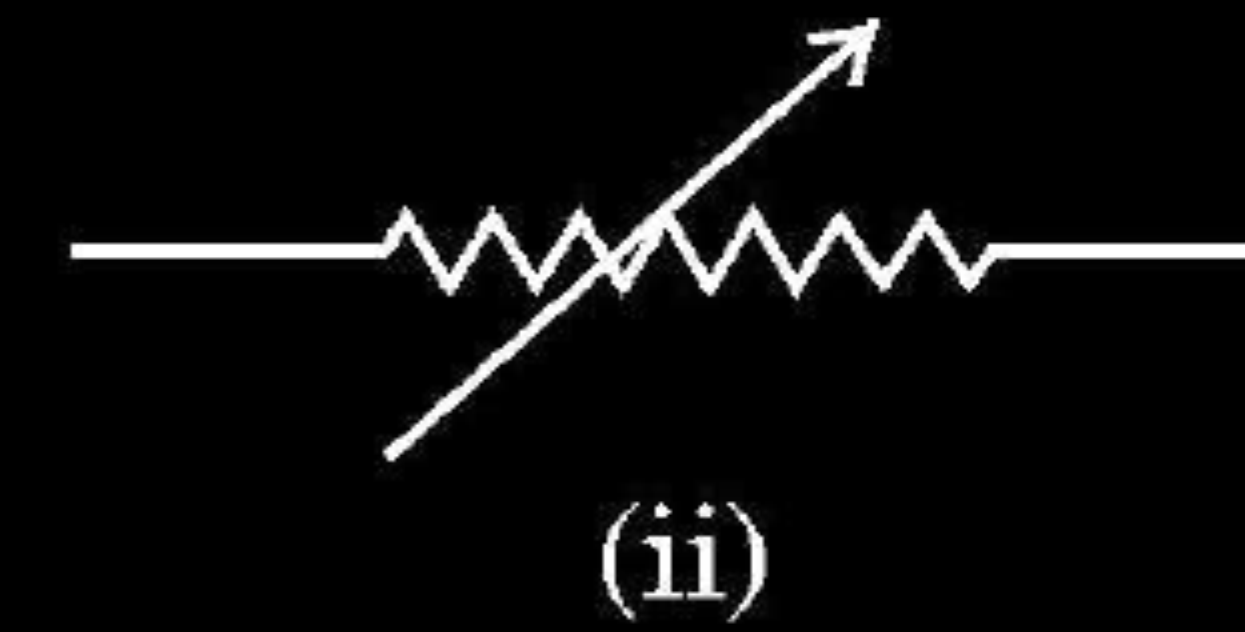
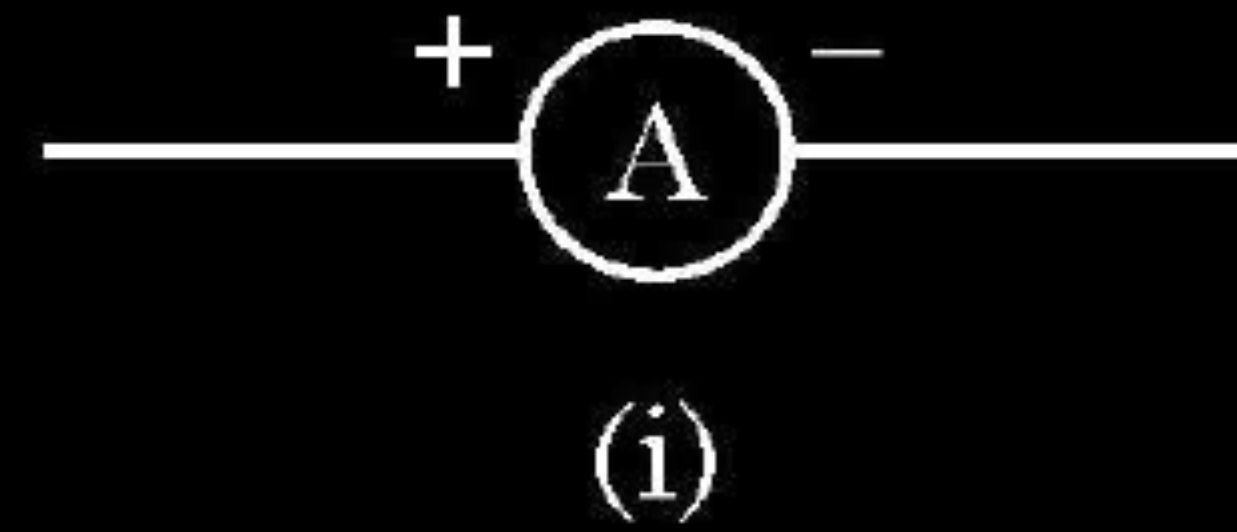
33. (a) Explain the statement “Potential difference between two points is 1 volt”.

(b) What do the symbols given below represent in an electric circuit ?

Write one function of each.

CBSE 2025

3



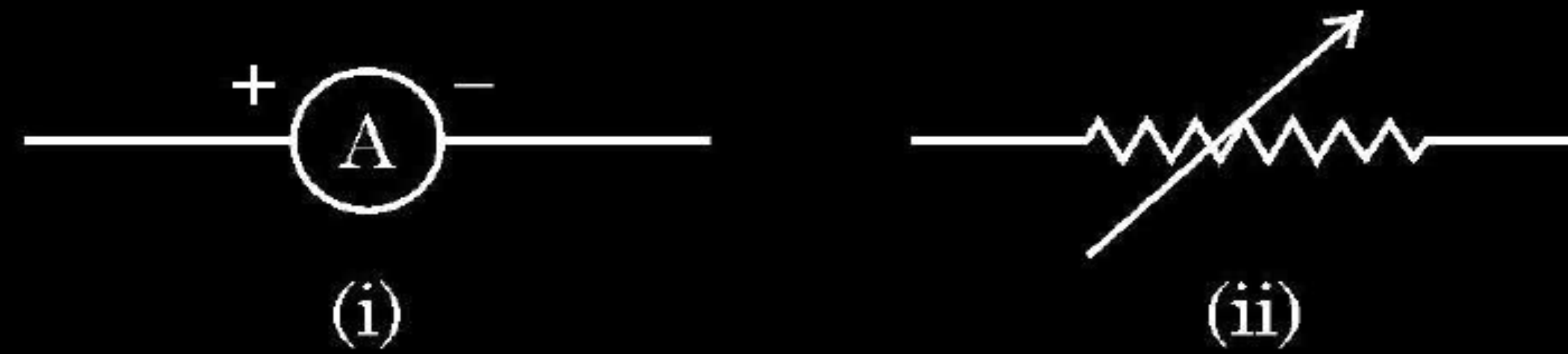
33. (a) Explain the statement “Potential difference between two points is 1 volt”.

(b) What do the symbols given below represent in an electric circuit ?

Write one function of each.

CBSE 2025

3



(a) It means 1 joule of work is done to move a charge of 1 coulomb from one point of the conductor to the other.

(b) (i)

Ammeter.

Ammeter is used to measure electric current

(ii)

Rheostat or variable resistance

Rheostat is used in a circuit to vary the resistance of the circuit

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2. In case of wires of same resistivity, the resistance of the wire will be maximum if the diameter and length of the wire respectively are

(A) D and L

(B) $\frac{D}{2}$ and $\frac{L}{4}$

(C) $\frac{D}{3}$ and $\frac{L}{6}$

(D) $\frac{D}{4}$ and $\frac{L}{12}$

CBSE 2025

2. In case of wires of same resistivity, the resistance of the wire will be maximum if the diameter and length of the wire respectively are

(A) D and L

(B) $\frac{D}{2}$ and $\frac{L}{4}$

(C) $\frac{D}{3}$ and $\frac{L}{6}$

(D) $\frac{D}{4}$ and $\frac{L}{12}$

CBSE 2025

(C)/ $D/3$ and $L/6$

28. The resistance of a wire of 0.01 cm radius is 7 ohms. If the resistivity of the material of the wire is 44×10^{-6} ohm meter, calculate the length of the wire.

CBSE 2025

28. The resistance of a wire of 0.01 cm radius is 7 ohms. If the resistivity of the material of the wire is 44×10^{-6} ohm meter, calculate the length of the wire.

CBSE 2025

$$r=0.01 \text{ cm}=0.01 \times 10^{-2} \text{ m}, R=7 \text{ ohm}$$

$$\rho=44 \times 10^{-6} \Omega \text{m}$$

$$A=\pi r^2$$

$$R = \rho \frac{l}{A} \quad \Rightarrow l = \frac{R \times A}{\rho}$$

$$l = \frac{7 \Omega \times (0.01 \times 10^{-2})^2 \times \frac{22}{7} \text{m}^2}{44 \times 10^{-6} \Omega \text{m}}$$

$$l = 0.5 \times 10^{-2} \text{m}$$

1/2

1/2

1/2

1

1/2

34. (a) (i) Write the relation between the electrical resistivity ' ρ ' and the resistance ' R ' of a cylindrical conductor of length ' l ' and area of cross-section ' A '. Use this relationship to determine the SI unit of ' ρ '.

CBSE 2024 C

- (ii) Find the electrical resistivity of the material of a wire of length 120 cm and diameter 0.4 mm. The resistance of this wire is $3\ \Omega$.

(a) (i)

$$\bullet \quad R = \rho \frac{l}{A}$$
$$\rho = \frac{R \times A}{l}$$

1

$$\bullet \quad \text{SI unit of } \rho = \frac{\text{SI unit of } R \times \text{SI unit of } A}{\text{SI unit of } l} = \frac{\Omega \times \text{m}^2}{\text{m}}$$

$$= \Omega \text{ m}$$

1

(ii) $R = 3\Omega$; $l = 120 \text{ cm} = 1.2 \text{ m}$;

$d = 0.4 \text{ mm}$ or $r = 0.2 \text{ mm}$

$\frac{1}{2}$

$$\rho = \frac{R \times A}{l}$$

$$A = 3.14 \times (0.2 \text{ mm})^2 = 3.14 \times 4 \times 10^{-8} \text{ m}^2$$

$\frac{1}{2}$

$$= 3\Omega \times \frac{3.14 \times 4 \times 10^{-8} \text{ m}^2}{1.2 \text{ m}}$$

1

$$= 3.14 \times 10^{-7} \Omega \text{ m}$$

1

36. (a) (i) State Ohm's law. Draw $V - I$ graph for an ohmic conductor.
- (ii) Electrical resistance of a metal wire of length 1 km is $200\ \Omega$. If the area of cross-section of the wire is $40 \times 10^{-8}\ \text{m}^2$, determine the electrical resistivity of the material of the wire.

CBSE 2024 C

36. (a) (i) State Ohm's law. Draw $V - I$ graph for an ohmic conductor.
- (ii) Electrical resistance of a metal wire of length 1 km is $200\ \Omega$. If the area of cross-section of the wire is $40 \times 10^{-8}\ \text{m}^2$, determine the electrical resistivity of the material of the wire.

CBSE 2024 C

5

(a)

(i)

- Ohm's law: The potential difference, V , across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it, provided its temperature remains the same. /

$$V \propto I$$

(ii) Here $l = 1\text{ km} = 1000\ \text{m}$

$R = 200\ \Omega$, $A = 40 \times 10^{-8}\ \text{m}^2$

$$\rho = \frac{R \times A}{l}$$

$$= 200\ \Omega \times \frac{40 \times 10^{-8}\ \text{m}^2}{1000\ \text{m}}$$

$$= 8 \times 10^{-8}\ \Omega\ \text{m}$$

1

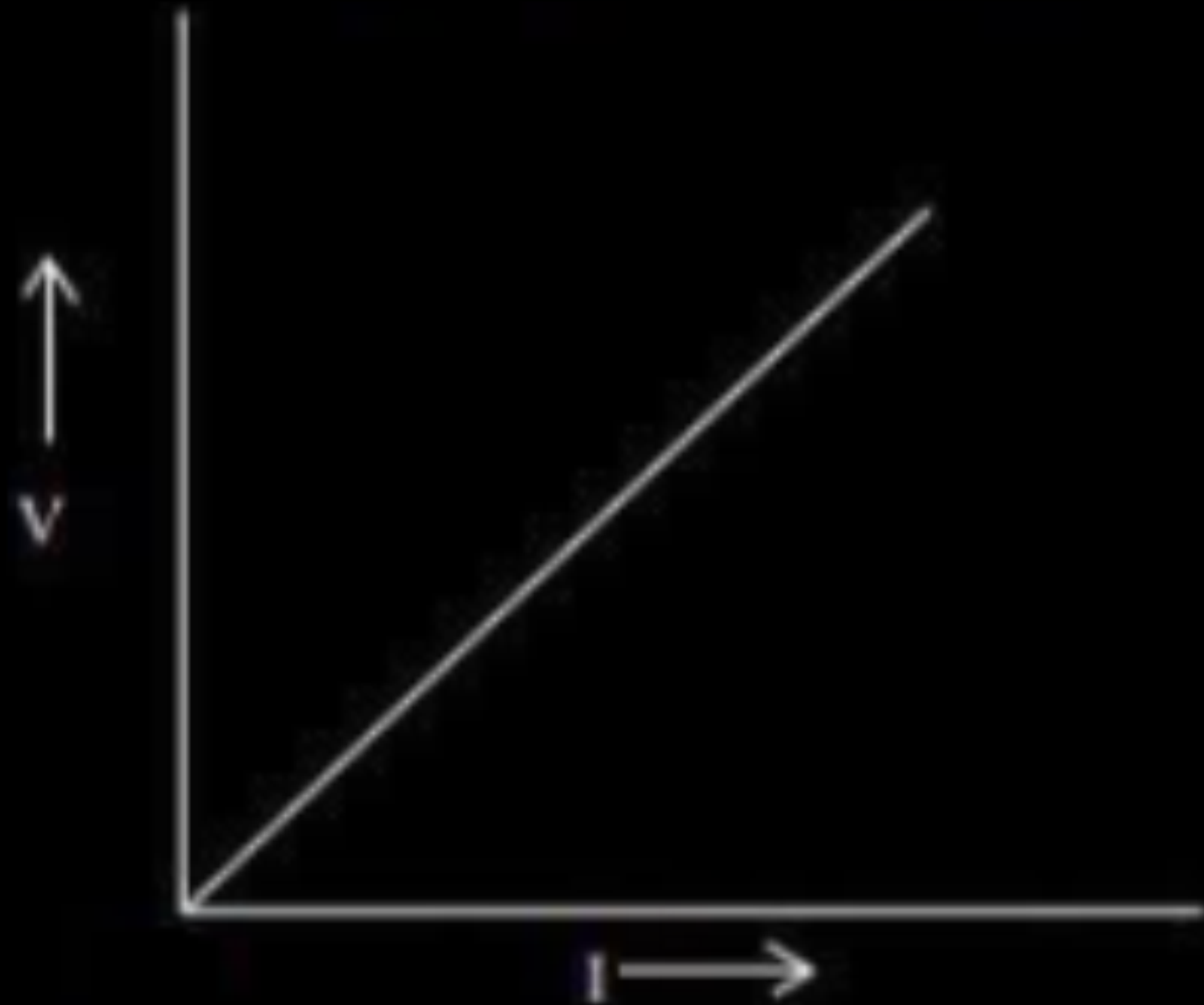
1

$\frac{1}{2}$

$\frac{1}{2}$

1

1



14. You have four resistance wires of same material. Select the wire which will offer the least resistance to the flow of current in an electric circuit :

- (A) Length $2l$ and diameter $2d$
- (B) Length l and diameter d
- (C) Length $l/2$ and diameter $d/2$
- (D) Length $l/4$ and diameter $d/4$

CBSE 2024 C

14. You have four resistance wires of same material. Select the wire which will offer the least resistance to the flow of current in an electric circuit :

CBSE 2024 C

- (A) Length $2l$ and diameter $2d$
- (B) Length l and diameter d
- (C) Length $l/2$ and diameter $d/2$
- (D) Length $l/4$ and diameter $d/4$

- 36.** (a) List two factors on which the resistance of a conductor in the shape of a straight cylinder depends. Write the relation between resistance (R) and the resistivity (ρ) of the material of the conductor. Use this relation to obtain the SI unit of resistivity.
- (b) Resistance of a metal wire of length 1 m is $40\ \Omega$. If the area of cross-section of the wire is $6.5 \times 10^{-8}\ \text{m}^2$, find the resistivity of the material of the wire.

(a) • Two factors (any two)

(i) Length (l)

(ii) area of cross-section (A)

(iii) Material of conductor

$$\bullet R = \rho \frac{l}{A} \quad \text{or} \quad \rho = R \frac{l}{A}$$

$$\bullet \text{SI unit of } \rho = \text{SI unit of (R)} \times \frac{\text{SI unit of (A)}}{\text{SI unit of (l)}}$$

$$= \text{ohm} \times \frac{\text{metre}^2}{\text{metre}} = \text{ohm} \cdot \text{metre}$$

$$(b) \rho = R \frac{l}{A}$$

$$= 40 \, \Omega \times \frac{6.5 \times 10^{-8} \, \text{m}^2}{1 \, \text{m}}$$

$$= 26 \times 10^{-7} \, \Omega \text{m}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$1$$

$$1$$

$$1$$

$$1$$

- 34.** (a) (i) Write the relation between resistivity (ρ) and resistance (R) for a uniform metallic conductor of length (l) and area of cross-section (A). Use this relation to obtain the SI unit of resistivity.
- (ii) The resistance of a metallic wire of 100 m is $20\ \Omega$. If the radius of the wire is 5 mm, find the resistivity of the metal of the wire.

CBSE 2024

$$(a) (i) \quad R = \rho \frac{l}{A} \Rightarrow \rho = \frac{RA}{l}$$

$$\text{SI unit of resistivity } (\rho) = \frac{\text{SI unit of } R \times \text{SI unit of } A}{\text{SI unit of } l}$$

$$= \frac{\text{ohm} \times (\text{metre})^2}{\text{metre}}$$

$$= \text{ohm metre or } \Omega \text{ m}$$

$$(ii) \quad l = 100 \text{ m}; R = 20 \Omega; r = 5 \text{ mm} = \frac{5}{1000} \text{ m}$$

$$\therefore A = \pi r^2 = 3.14 \times \left(\frac{5}{1000} \right)^2 \text{ m}^2$$

$$\therefore \rho = R \frac{A}{l}$$

$$= 20 \Omega \times \frac{3.14 \times \left(\frac{5}{1000} \right)^2 \text{ m}^2}{100 \text{ m}}$$

$$= 157 \times 10^{-7} \Omega \text{ m}$$

1

1/2

1/2

1

1

1

17. *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

CBSE 2024

17. *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

CBSE 2024

(B) / Both Assertion (A) and Reason (R) are the true , but Reason (R) is not a correct explanation of Assertion (A).

26. The filament of an electric lamp draws a current of 0.5 A , which lights for 2 hours. Calculate the charge that flows through the circuit.

CBSE 2024

2

- 26.** The filament of an electric lamp draws a current of 0.5 A, which lights for 2 hours. Calculate the charge that flows through the circuit. CBSE 2024

2

$$I = 0.5 \text{ A}, t = 2 \text{ hours} = 2 \times 3600 \text{ s}$$

$$I = \frac{Q}{t}$$

$$\therefore Q = I \times t = 0.5 \text{ A} \times 2 \times 3600 \text{ s} = 3600 \text{ C}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

- (b) The values of current I flowing in a given resistor for the corresponding values of potential difference V across the resistor are given in the following table :

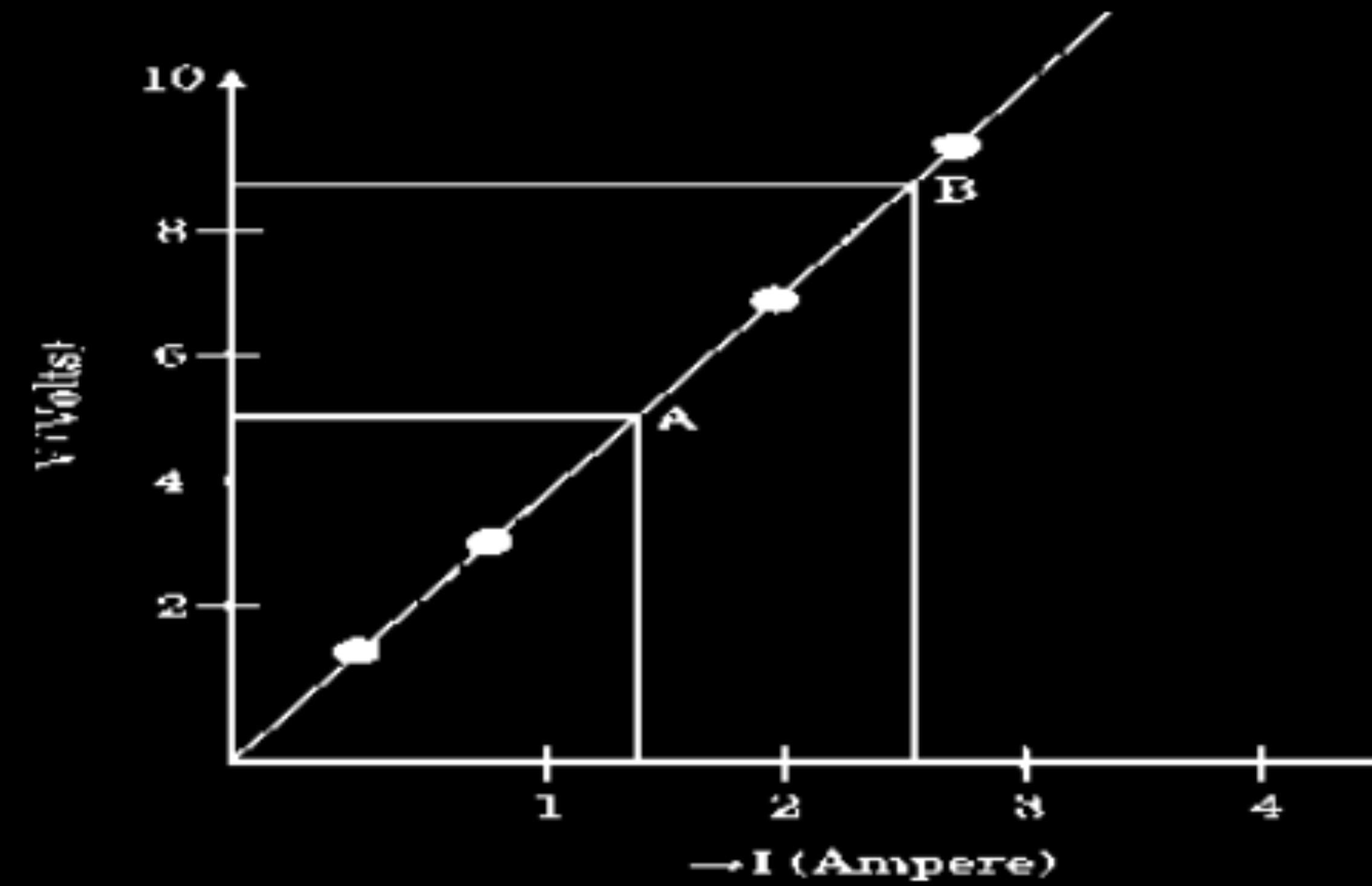
I (Amperes)	0.5	1.0	2.0	3.0	3.5
V (Volts)	1.5	3.0	6.2	9.3	10.8

- (i) Plot a graph between V and I .
- (ii) Calculate the resistance of the resistor with the help of the graph.
- (iii) What does the graph represent ?
- (iv) Why should this graph pass through the origin ?

CBSE 2024

(b)

(i)



(ii) Resistance of resistor = $\frac{V_2 - V_1}{I_2 - I_1} = \frac{8.3 - 5.2}{2.5 - 1.5} = 3.1 \Omega$

(iii) The given resistor obeys Ohm's law./ Resistance remains constant.

(iv) Because when the value of $V = 0$, the current $I = 0$.

1

1+1

1

1

- 26.** An electric source can supply a charge of 750 coulomb. If the current drawn by a device is 15 mA, find the time in which the electric source will be discharged completely.

CBSE 2024

2

26. An electric source can supply a charge of 750 coulomb. If the current drawn by a device is 15 mA, find the time in which the electric source will be discharged completely.

CBSE 2024

2

- | | |
|---|---|
| • $Q = I \times t \Rightarrow t = \frac{Q}{I}$ | 1 |
| • $\therefore t = \frac{750}{\frac{15}{1000}} = \frac{750 \times 1000}{15} = 50000 \text{ s}$ | 1 |

- (b) (i) When is the potential difference between two points said to be 1 volt ?
- (ii) A copper wire has a diameter of 0.2 mm and resistivity of $1.6 \times 10^{-8} \Omega \text{ m}$. What will be the length of this wire to make its resistance 14Ω ? How much does the resistance change, if the diameter of the wire is doubled ?

CBSE 2024

(b)

- (i) When 1 joule of work is done to move a charge of 1 coulomb from one point to the other.

$$d = 0.2 \text{ mm} = 2 \times 10^{-4} \text{ m}; R = 14 \Omega$$

$$\rho = 1.6 \times 10^{-8} \Omega \text{ m}; A = \frac{\pi d^2}{4}$$

$$R = \frac{\rho l}{A} = \frac{4\rho l}{\pi d^2} \text{ or } l = \frac{\pi d^2 R}{4\rho}$$

$$l = \frac{22}{7} \times \frac{(2 \times 10^{-4})^2}{4 \times 1.6 \times 10^{-8}} \times 14$$
$$= \frac{22 \times 14}{7 \times 1.6} = 27.5 \text{ m}$$

When the diameter is doubled, $d' = 2d$

$$A' = 4A$$

$$\frac{R'}{R} = \frac{A}{A'} \text{ or } R' = \frac{RA}{A'} = \frac{RA}{4A}$$

$$R' = \frac{R}{4} = \frac{14 \Omega}{4} = 3.5 \Omega$$

$$\text{Change } (14.0 - 3.5) = 10.5 \Omega$$

1

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

1

$\frac{1}{2}$

26. Calculate the resistance of a copper wire of length 1000 m and area of cross-section 2 mm^2 . Resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$.

CBSE 2024

2

26. Calculate the resistance of a copper wire of length 1000 m and area of cross-section 2 mm^2 . Resistivity of copper is $1.6 \times 10^{-8} \Omega \text{ m}$.

CBSE 2024

2

$$R = \rho \frac{l}{A}$$

$$\rho = 1.6 \times 10^{-8} \Omega \text{ m}$$

$$A = 2 \times (10^{-3} \text{ m})^2$$

$$l = 1000 \text{ m}$$

$$\begin{aligned} \therefore R &= (1.6 \times 10^{-8} \Omega \text{ m}) \times \frac{1000 \text{ m}}{2 \times (10^{-3} \text{ m})^2} \\ &= 8.0 \Omega \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

- 26.** Two wires A and B of same material, having same lengths and diameters 0·2 mm and 0·3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

CBSE 2024

2

26. Two wires A and B of same material, having same lengths and diameters 0.2 mm and 0.3 mm respectively, are connected one by one in a circuit. Which one of these two wires will offer more resistance to the flow of current in the circuit ? Justify your answer.

CBSE 2024

2

- Wire A will offer more resistance

Justification:

- $R \propto \frac{l}{A}$ /thinner the wire, more resistance to the flow of current.

1

1

13. In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are

1

(A) $D/2$ and $L/4$

(B) $D/4$ and $4L$

(C) $2D$ and L

(D) $4D$ and $2L$

CBSE 2024

13. In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are

1

(A) $D/2$ and $L/4$

(B) $D/4$ and $4L$

(C) $2D$ and L

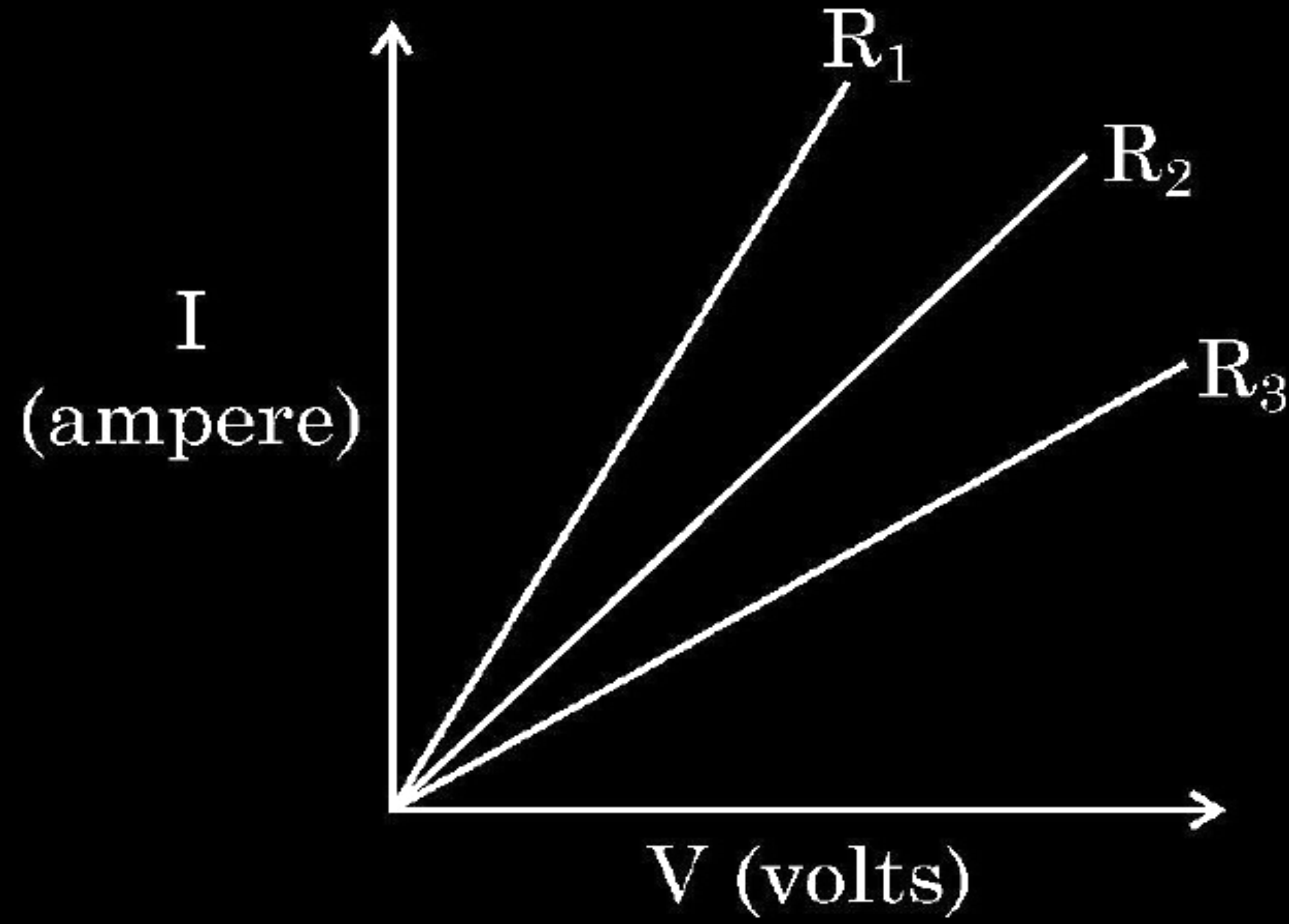
(D) $4D$ and $2L$

CBSE 2024

(D) / $4D$ and $2L$

13. Study the I-V graph for three resistors of resistances R_1 , R_2 and R_3 and select the correct statement from the following :

1



CBSE 2024

(A) $R_1 = R_2 = R_3$

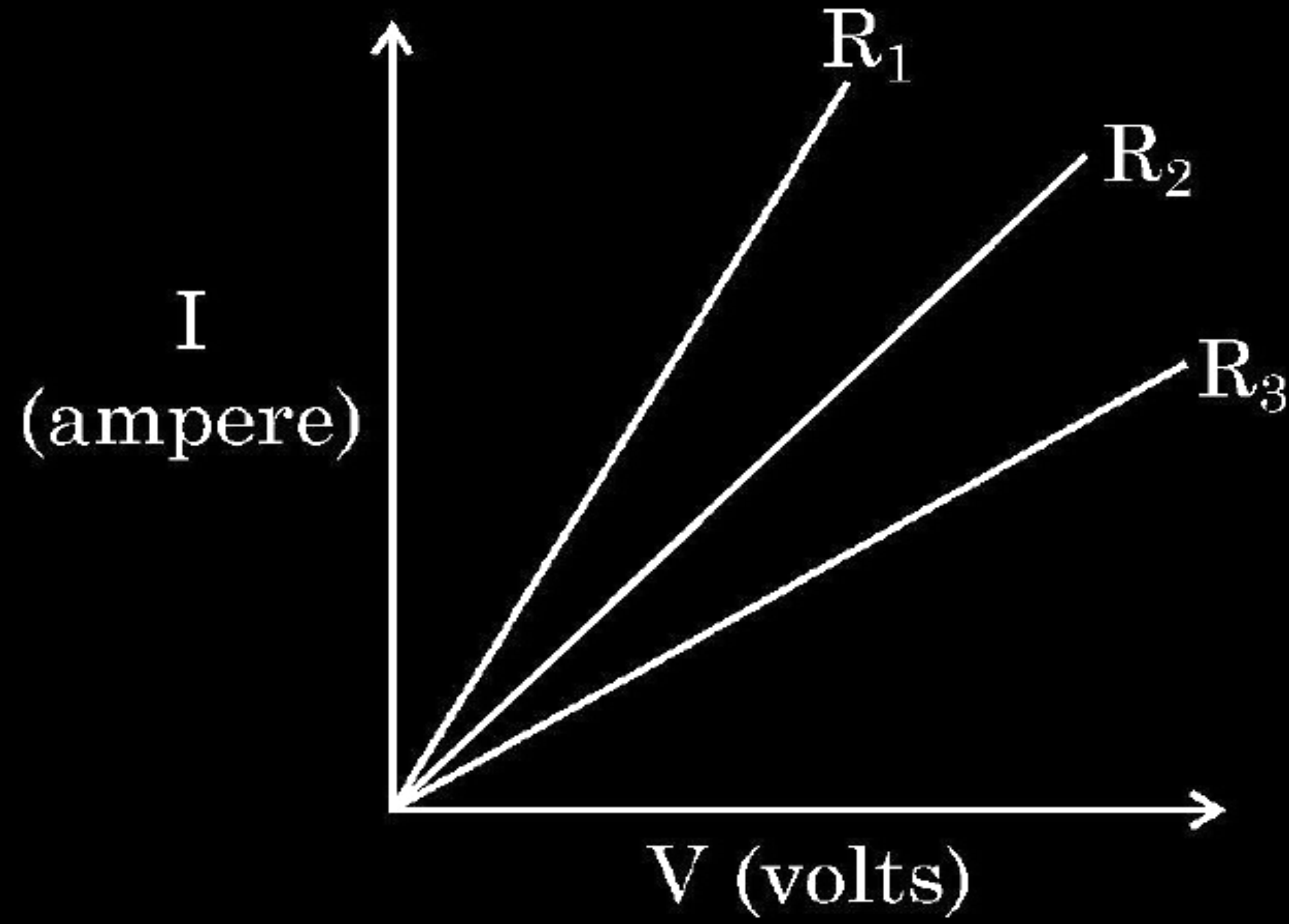
(B) $R_1 > R_2 > R_3$

(C) $R_3 > R_2 > R_1$

(D) $R_2 > R_3 > R_1$

13. Study the I-V graph for three resistors of resistances R_1 , R_2 and R_3 and select the correct statement from the following :

1



CBSE 2024

- (A) $R_1 = R_2 = R_3$ (B) $R_1 > R_2 > R_3$
(C) $R_3 > R_2 > R_1$ (D) $R_2 > R_3 > R_1$

(C) $R_3 > R_2 > R_1$

14. S.I. unit of electrical resistivity is

1

(A) ohm per metre³

(B) ohm per metre²

(C) ohm . metre

(D) ohm . metre³

CBSE 2024

14. S.I. unit of electrical resistivity is

1

(A) ohm per metre³

(B) ohm per metre²

(C) ohm . metre

(D) ohm . metre³

CBSE 2024

(C) /ohm.metre

- (B)** (i) Write the relation between resistance R and electrical resistivity ρ of the material of a conductor in the shape of cylinder of length l and area of cross-section A . Hence derive the SI unit of electrical resistivity.
- (ii) The resistance of a metal wire of length 3 m is $60\ \Omega$. If the area of cross-section of the wire is $4 \times 10^{-7}\ \text{m}^2$, calculate the electrical resistivity of the wire.
- (iii) State how would electrical resistivity be affected if the wire (of part 'ii') is stretched so that its length is doubled. Justify your answer.

(B)

(i) $R = \rho \frac{l}{A}$

1

$$\rho = \frac{R \times A}{l}$$
$$= \text{ohm} \times \frac{(\text{metre})^2}{\text{metre}}$$

$$= \text{ohm meter} / \Omega\text{m}$$

1

(ii) Here $l = 3 \text{ m}$, $A = 4 \times 10^{-7} \text{ m}^2$, $R = 60 \Omega$

$$\rho = \frac{R \times A}{l}$$
$$= \frac{60 \times 4 \times 10^{-7}}{3}$$
$$= 80 \times 10^{-7} \Omega\text{m}$$

1

(iii)

- Resistivity will not change.
- because Resistivity does not depend on the dimension of the conductor / It only depends on the nature of the material.

1

1

11. The resistance of a wire does ***not*** depend on its :

- (a) Length
- (b) Area of cross-section
- (c) Shape
- (d) Material

CBSE 2023 C

11. The resistance of a wire does ***not*** depend on its :

- (a) Length
- (b) Area of cross-section
- (c) Shape
- (d) Material

CBSE 2023 C

14. How much work is done in moving a charge of 3 C across two points having a potential difference of 12 V ?

- (a) 4 J
- (b) 36 J
- (c) 54 J
- (d) 216 J

CBSE 2023 C

14. How much work is done in moving a charge of 3 C across two points having a potential difference of 12 V ?

- (a) 4 J
- (b) 36 J
- (c) 54 J
- (d) 216 J

CBSE 2023 C

(b)

9. A material used for making heating elements of electrical heating devices should have :

- (a) High resistivity and high melting point.
- (b) High resistivity and low melting point.
- (c) Low resistivity and high melting point.
- (d) Low resistivity and low melting point.

CBSE 2023 C

9. A material used for making heating elements of electrical heating devices should have :

- (a) High resistivity and high melting point.
- (b) High resistivity and low melting point.
- (c) Low resistivity and high melting point.
- (d) Low resistivity and low melting point.

CBSE 2023 C

- 36.** (a) Calculate the resistance of an aluminium wire of length 1 m and area of cross-section 2 mm^2 . Resistivity of aluminium is $2.63 \times 10^{-8} \Omega\text{m}$.
- (b) From the values given below, plot a graph of I versus V. Show that the data is in conformity of the Ohm's law.

5

Current – I (ampere)	0.1	0.2	0.3	0.4
Potential Difference – V (volt)	1.2	2.4	3.6	4.8

CBSE 2023 C

$$(a) \ l = 1 \text{ m}$$

$$A = 2\text{mm}^2 = 2 \times (10^{-3}\text{m})^2 = 2 \times 10^{-6} \text{ m}^2$$

$$\rho = 2.63 \times 10^{-8} \ \Omega\text{m}$$

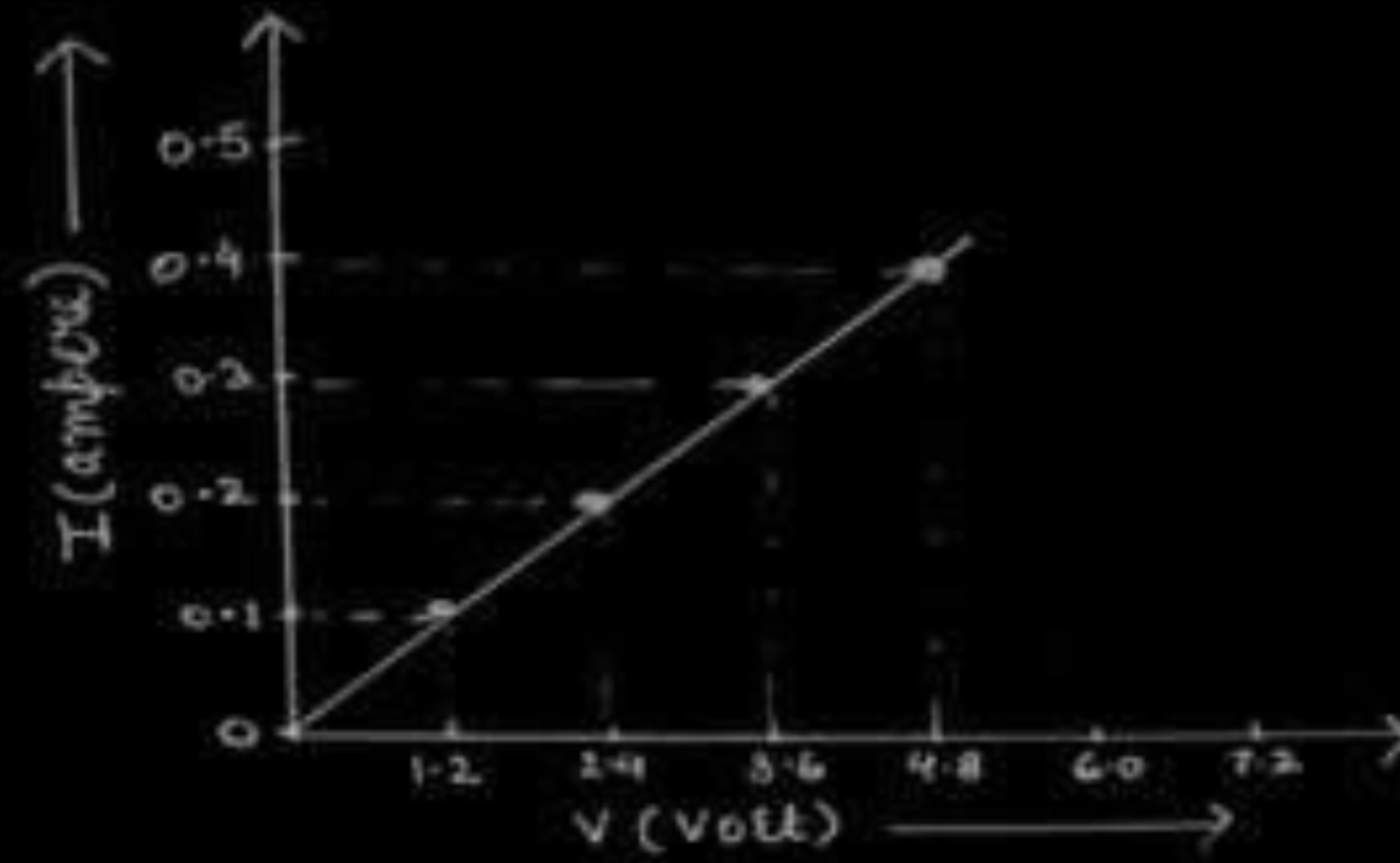
$$R = \rho \frac{l}{A}$$

$$R = (2.63 \times 10^{-8} \ \Omega\text{m}) \times \frac{1\text{m}}{2 \times 10^{-6} \text{ m}^2}$$

$$= \frac{2.63}{2} \times 10^{-2}$$

$$= 1.315 \times 10^{-2} \Omega$$

(



Since I – V graph is a straight line passing through the origin it shows that $I \propto V$ which is in accordance with the Ohm's law.

13. Potential difference can also be represented as :

(a)
$$\frac{\text{Work done}}{\text{Charge}}$$

(b)
$$\text{Work done} \times \text{Charge}$$

(c)
$$\text{Work done} \times \text{Charge} \times \text{Time}$$

(d)
$$\frac{\text{Work done} \times \text{Charge}}{\text{Time}}$$

CBSE 2023 C

13. Potential difference can also be represented as :

(a) $\frac{\text{Work done}}{\text{Charge}}$

(b) $\text{Work done} \times \text{Charge}$

(c) $\text{Work done} \times \text{Charge} \times \text{Time}$

(d) $\frac{\text{Work done} \times \text{Charge}}{\text{Time}}$

CBSE 2023 C

(a)

- 32.** (a) State Ohm's law. List two factors on which the resistance of a cylindrical conductor at a given temperature depends. A wire was drawn into a wire of double its length by melting it. If the resistance of the old wire was R , what is the resistance of the new wire ?

CBSE 2023 C

32. (a) State Ohm's law. List two factors on which the resistance of a cylindrical conductor at a given temperature depends. A wire was drawn into a wire of double its length by melting it. If the resistance of the old wire was R , what is the resistance of the new wire ? **CBSE 2023 C**

3

13. SI unit of electrical resistivity is :

(a) ohm metre

(b) ohm per metre

(c) ohm / metre²

(d) ohm / metre³

CBSE 2023

13. SI unit of electrical resistivity is :

- | | |
|------------------------------|------------------------------|
| (a) ohm metre | (b) ohm per metre |
| (c) ohm / metre ² | (d) ohm / metre ³ |

CBSE 2023

—
(a)
—

14. You have a thin resistance wire having a thick plastic insulation. A decrease in the resistance of this wire would be produced if somehow we increase the :

- (a) thickness of the plastic insulation
- (b) temperature of the wire
- (c) length of the wire
- (d) diameter of the wire

CBSE 2023

14. You have a thin resistance wire having a thick plastic insulation. A decrease in the resistance of this wire would be produced if somehow we increase the :

- (a) thickness of the plastic insulation
- (b) temperature of the wire
- (c) length of the wire
- (d) diameter of the wire

CBSE 2023

(d)

21. Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it ? State the law which helps us in solving the above stated question.

CBSE 2023

2

21. Let the resistance of an electrical device remain constant, while the potential difference across its two ends decreases to one fourth of its initial value. What change will occur in the current through it ? State the law which helps us in solving the above stated question.

CBSE 2023

2

- Current becomes one fourth of its original value.
- According to the Ohm's law – potential difference is directly proportional to the current flowing through the conductor provided temperature remains constant. /

$$V \propto I \text{ (Temperature remaining constant for a given conductor)}$$

1

1

34. (a) (i) What is meant by resistance of a conductor ? Define its SI unit.
- (ii) List two factors on which the resistance of a rectangular conductor depends.
- (iii) How will the resistance of a wire be affected if its
- (1) length is doubled, and
 - (2) radius is also doubled ?

Give justification for your answer.

CBSE 2023

(a) (i) The property of conductor to resist the flow of charges through it.	1
If Potential difference across the two ends of a conductor is 1V and the current through it is 1A, then resistance 'R' of the conductor is 1Ω.	1
Alternate answer $1\Omega = \frac{1 \text{ volt}}{1 \text{ ampere}} \quad / \quad \frac{1 \text{ V}}{1\text{A}}$	
(ii) • Length of the conductor • Area of cross-section of the conductor • Nature of the material • Temperature (any two)	$\frac{1}{2}$, $\frac{1}{2}$
(iii) (1) The resistance will become one half of its original value.	
$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2}$	$\frac{1}{2}$
$R' = \frac{\rho \cdot 2L}{\pi (2r)^2}$	$\frac{1}{2}$
(2) $R' = \frac{\rho \cdot l}{\pi (2r)^2} \cdot \frac{2}{4} = \frac{R}{2}$	$\frac{1}{2}$
Resistance will reduce to one half.	$\frac{1}{2}$

14. The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are (Here the symbols have their usual meanings) :

CBSE 2023

(a) (i) $I = \frac{Q}{t}$ (ii) $W = \frac{V}{Q}$

(b) (i) $Q = I \times t$ (ii) $W = V \times Q$

(c) (i) $Q = \frac{I}{t}$ (ii) $V = \frac{W}{Q}$

(d) (i) $I = \frac{Q}{t}$ (ii) $Q = \frac{V}{W}$

14. The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are (Here the symbols have their usual meanings) :

CBSE 2023

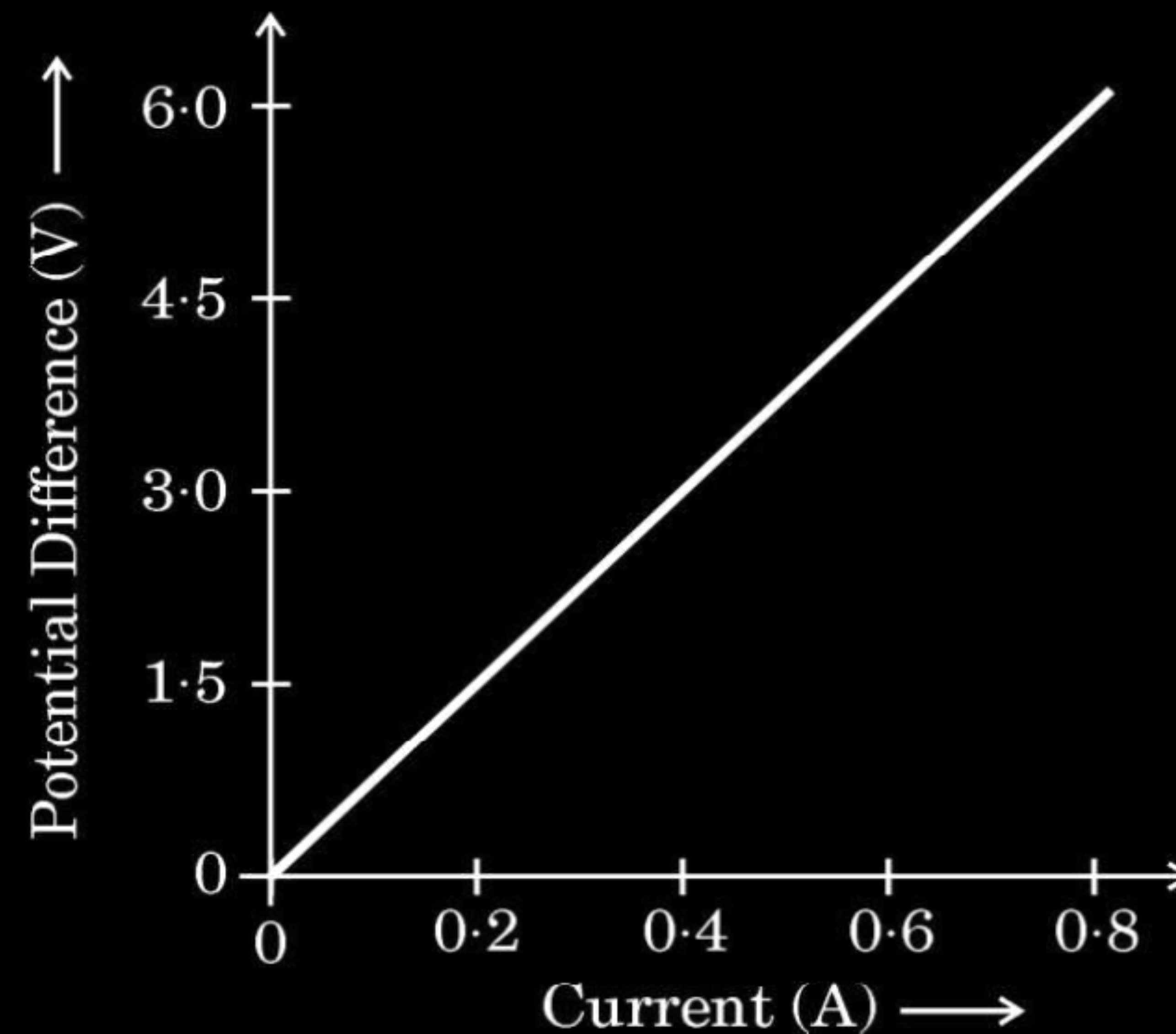
(a) (i) $I = \frac{Q}{t}$ (ii) $W = \frac{V}{Q}$

(b) (i) $Q = I \times t$ (ii) $W = V \times Q$

(c) (i) $Q = \frac{I}{t}$ (ii) $V = \frac{W}{Q}$

(d) (i) $I = \frac{Q}{t}$ (ii) $Q = \frac{V}{W}$

- 36.** (a) State Ohm's Law.
- (b) Name and define the physical quantity determined by the slope of $V - I$ curve given in the diagram. Use this graph to find the value of this physical quantity in SI units.



CBSE 2023

- (c) Establish the relationship between 1 kWh and 1 joule.

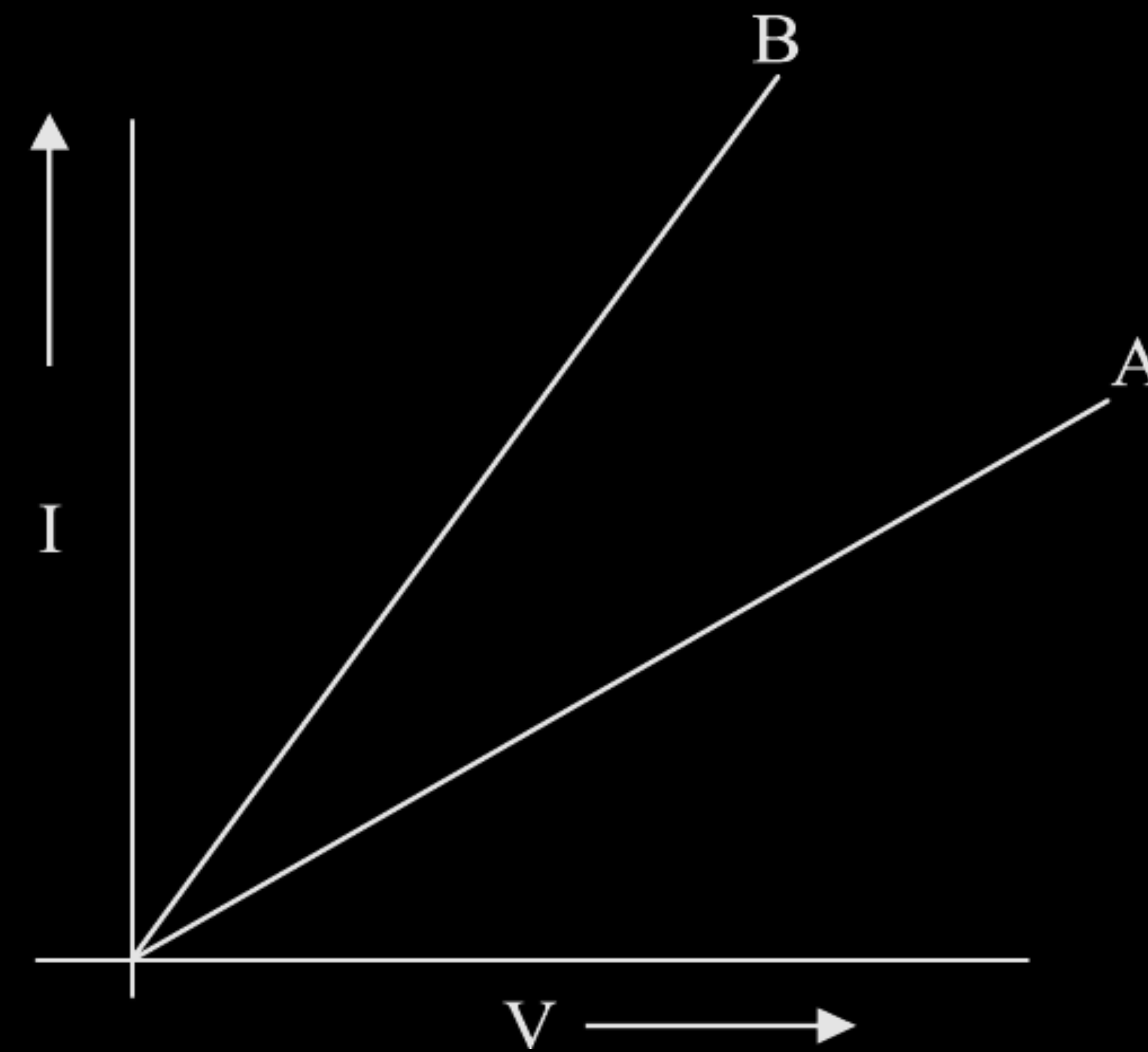
(a) The potential difference V across the ends of a given metallic wire in an electric circuit is directly proportional to the current flowing through it provided its temperature remains the same. / $V \propto I$ or $V = IR$ (temperature remaining constant)	1
(b) Resistance Resistance is the property of a conductor to resist the flow of charges. $R = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} / \frac{V}{I} = \frac{6 - 0}{0.8 - 0}$ $= \frac{60}{8} = 7.5 \, \Omega$	1 1 $\frac{1}{2}$ $\frac{1}{2}$
(c) $1 \text{ kWh} = 1000 \frac{J}{\text{sec}} \times 3600 \text{ sec}$ $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$	1

35. (i) How is electric current related to the potential difference across the terminals of a conductor ? **5**

Draw a labelled circuit diagram to verify this relationship.

(ii) Why should an ammeter have low resistance ?

(iii) Two V - I graphs A and B for series and parallel combinations of two resistors are as shown. Giving reason state which graph shows (a) series, (b) parallel combination of the resistors.



CBSE 2023

8. For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the :

- (a) ammeter is connected in parallel and the voltmeter in series.
- (b) ammeter is connected in series and the voltmeter in parallel.
- (c) ammeter and voltmeter are both connected in series.
- (d) ammeter and voltmeter are both connected in parallel.

CBSE 2023

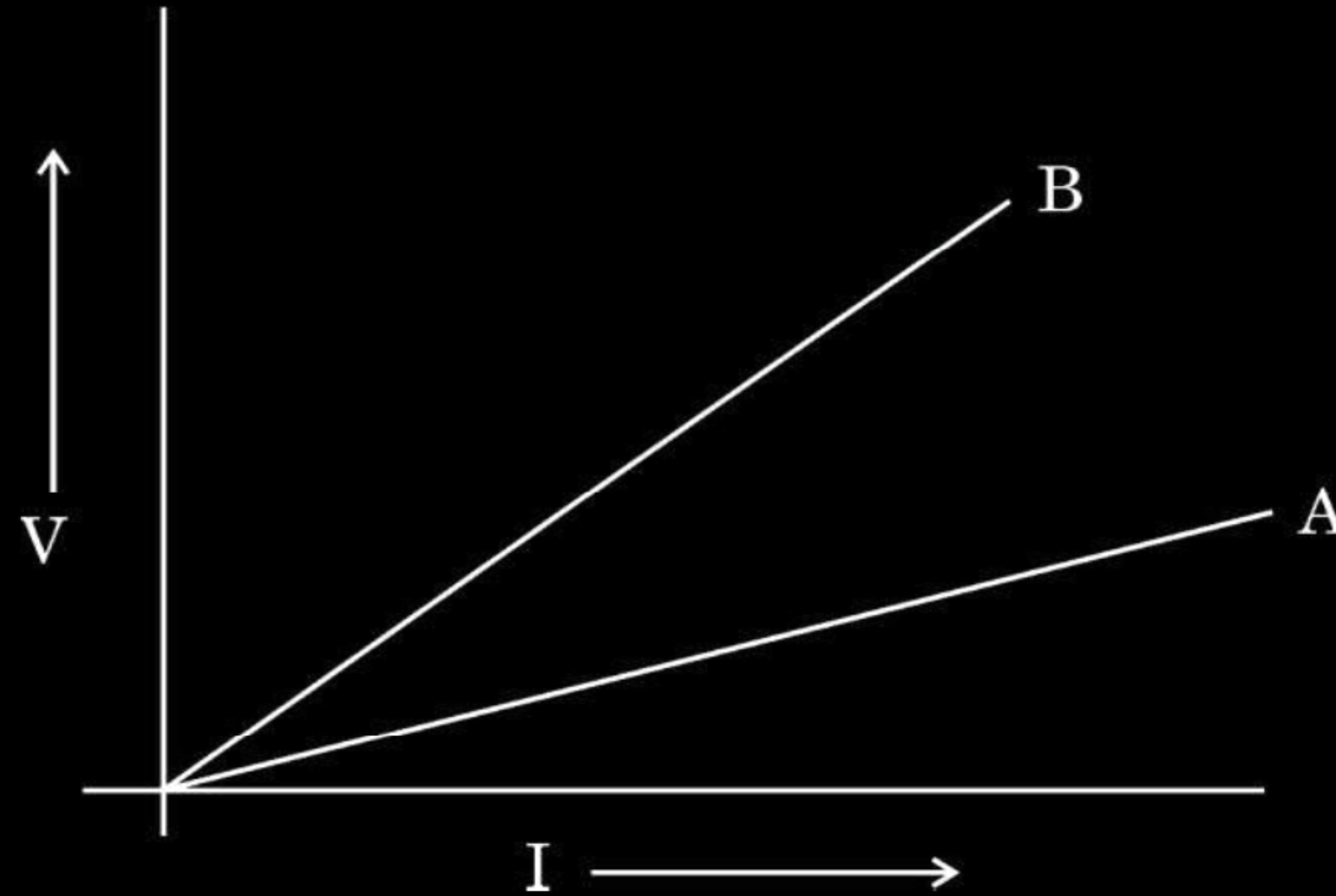
8. For verifying Ohm's law, we design an electric circuit diagram in which we show the arrangement of different circuit components. We find that with respect to the resistor, the :

- (a) ammeter is connected in parallel and the voltmeter in series.
- (b) ammeter is connected in series and the voltmeter in parallel.
- (c) ammeter and voltmeter are both connected in series.
- (d) ammeter and voltmeter are both connected in parallel.

CBSE 2023

21. $V - I$ graph for two conducting wires A and B are as shown. If both wires are of the same length and same diameter, which of the two is made of a material of high resistivity ? Give reasons to justify your answer.

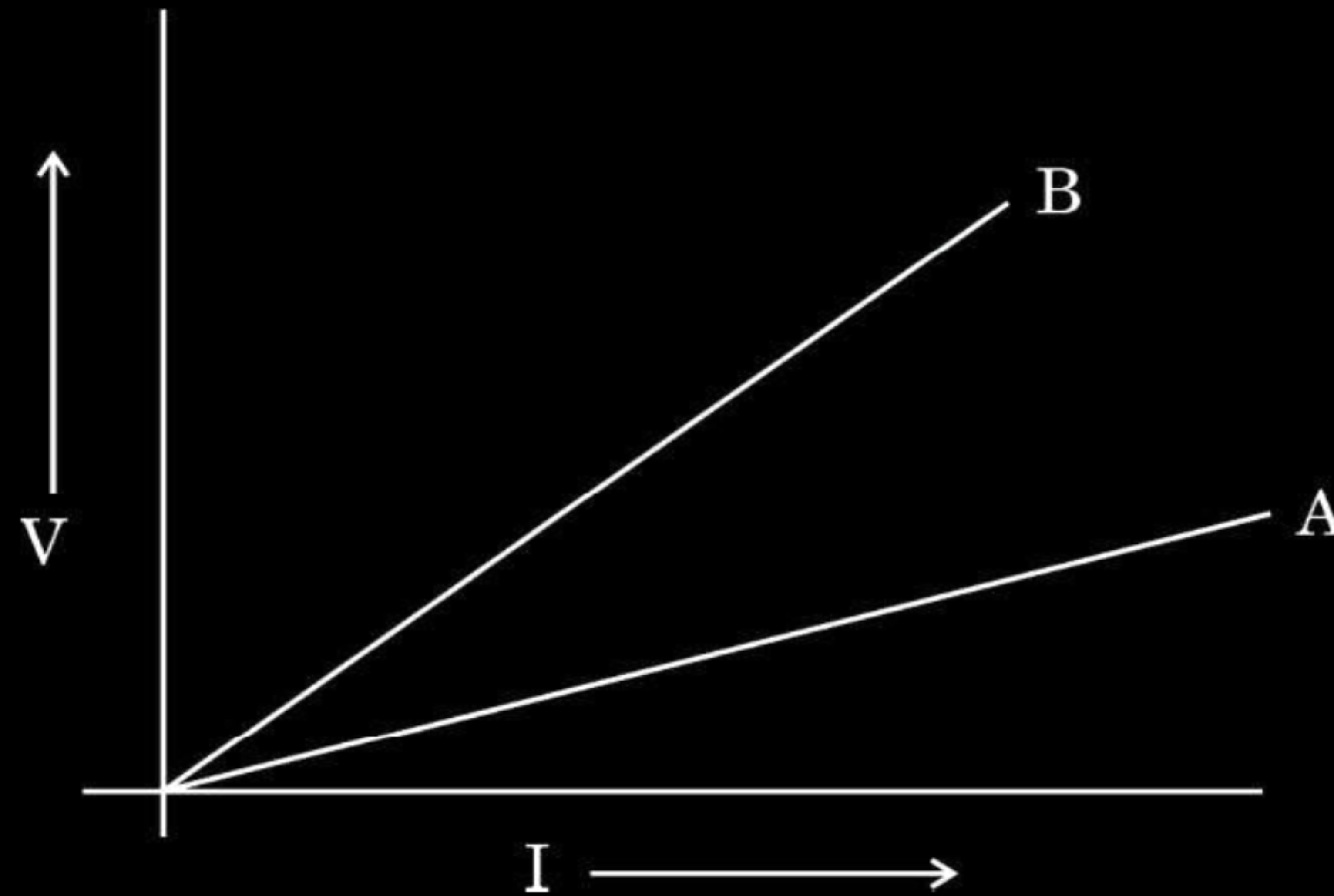
2



CBSE 2023

21. $V - I$ graph for two conducting wires A and B are as shown. If both wires are of the same length and same diameter, which of the two is made of a material of high resistivity ? Give reasons to justify your answer.

2



CBSE 2023

- Wire B.
- For the conductors of the same dimensions greater the resistance, greater is the resistivity.

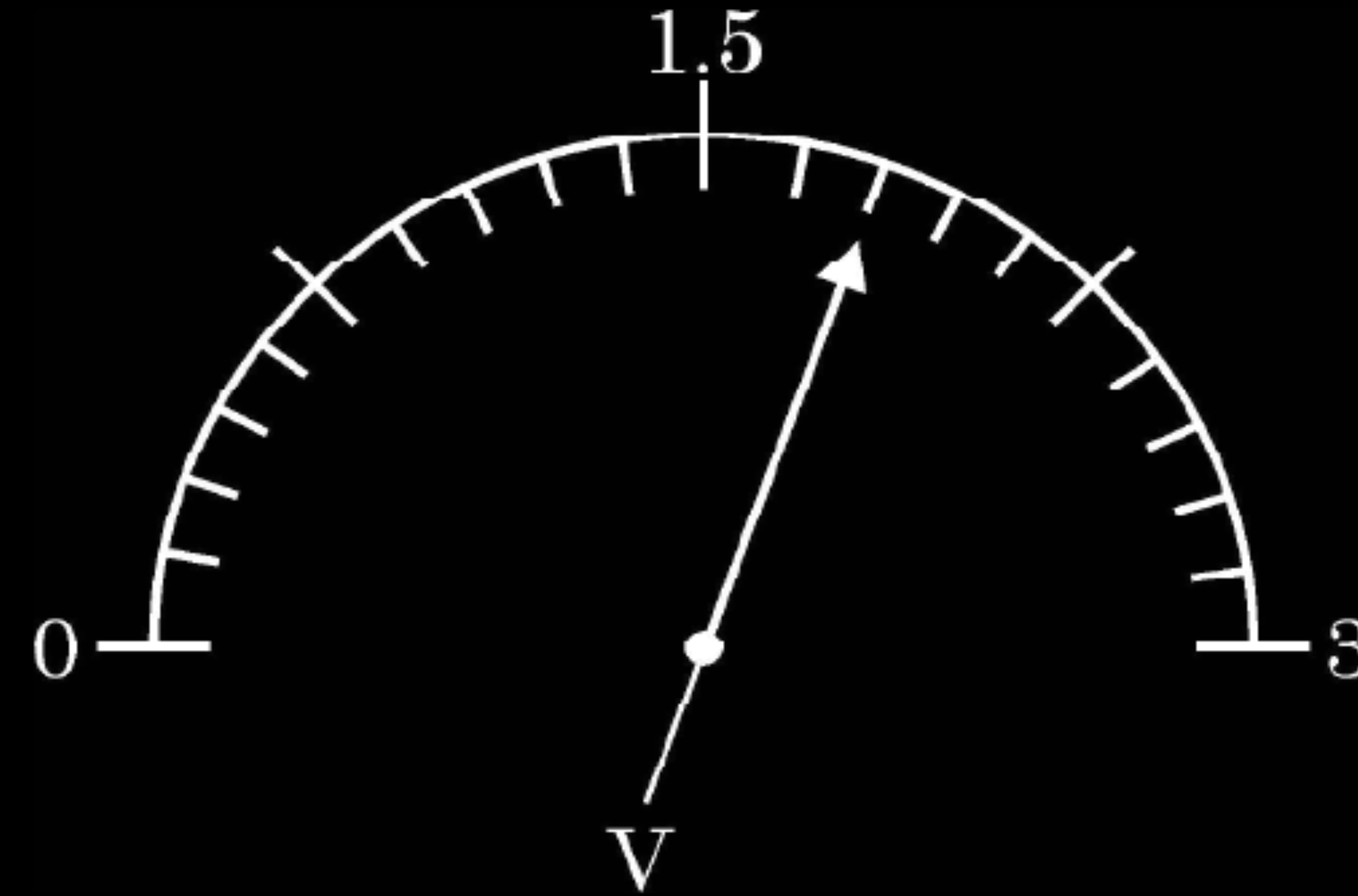
1

1

15. In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are :

CBSE 2023

1



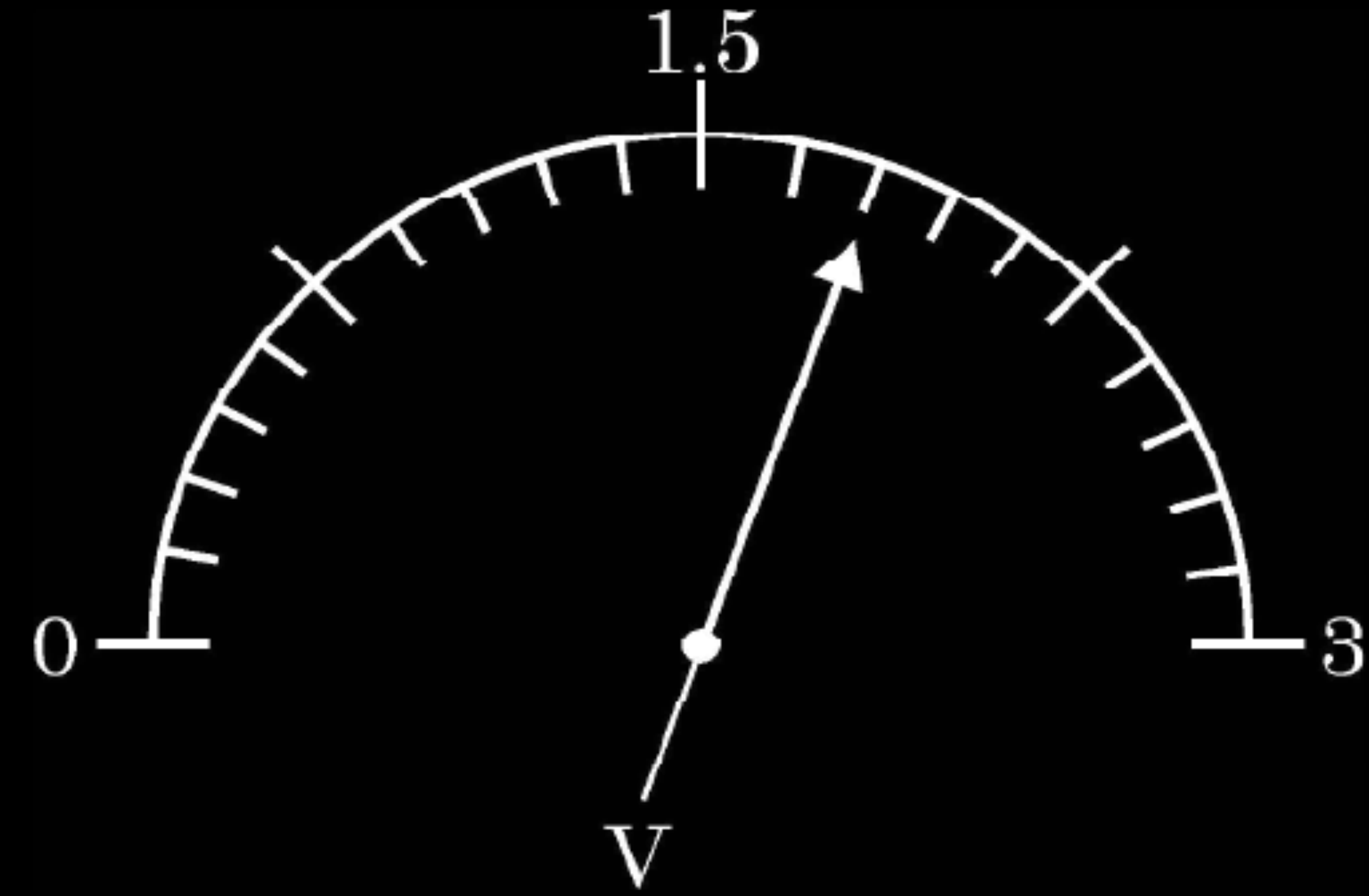
- (a) 0.15 V and 1.6 V
(c) 0.15 V and 1.8 V

- (b) 0.05 V and 1.6 V
(d) 0.05 V and 1.8 V

15. In the following diagram, the position of the needle is shown on the scale of a voltmeter. The least count of the voltmeter and the reading shown by it respectively are :

CBSE 2023

1



- (a) 0.15 V and 1.6 V
(c) 0.15 V and 1.8 V

- (b) 0.05 V and 1.6 V
(d) 0.05 V and 1.8 V

(c)

- 6.** (a) State Ohm's law. A student has drawn $V - I$ graph on the basis of his experimental observations. Why should this graph be a straight line passing through the origin ? 2

OR

- (b) When a 6.0 V battery is connected across a resistor there is a current of 30 mA in the circuit. Find the value of the resistance of the resistor in ohms. 2

CBSE 2022 C

6. (a) State Ohm's law. A student has drawn $V - I$ graph on the basis of his experimental observations. Why should this graph be a straight line passing through the origin ? 2

OR

- (b) When a 6.0 V battery is connected across a resistor there is a current of 30 mA in the circuit. Find the value of the resistance of the resistor in ohms. CBSE 2022 C 2

(a) Current flowing through a given conductor is directly proportional to potential difference applied across the conductor provided that temperature remains constant. / $V \propto I$ (when T is constant)

• According to Ohm's law, there should be no current in conductor if potential difference is zero. That is why, this graph be straight line passing through the origin.

OR

(b) $V=6.0 \text{ V}$ $I=30 \text{ mA}$

$$R = \frac{V}{I}$$

$$R = \frac{6.0 \text{ V}}{30 \times 10^{-3} \text{ A}}$$

$$R = 200 \, \Omega$$

1

1

$\frac{1}{2}$

1

$\frac{1}{2}$

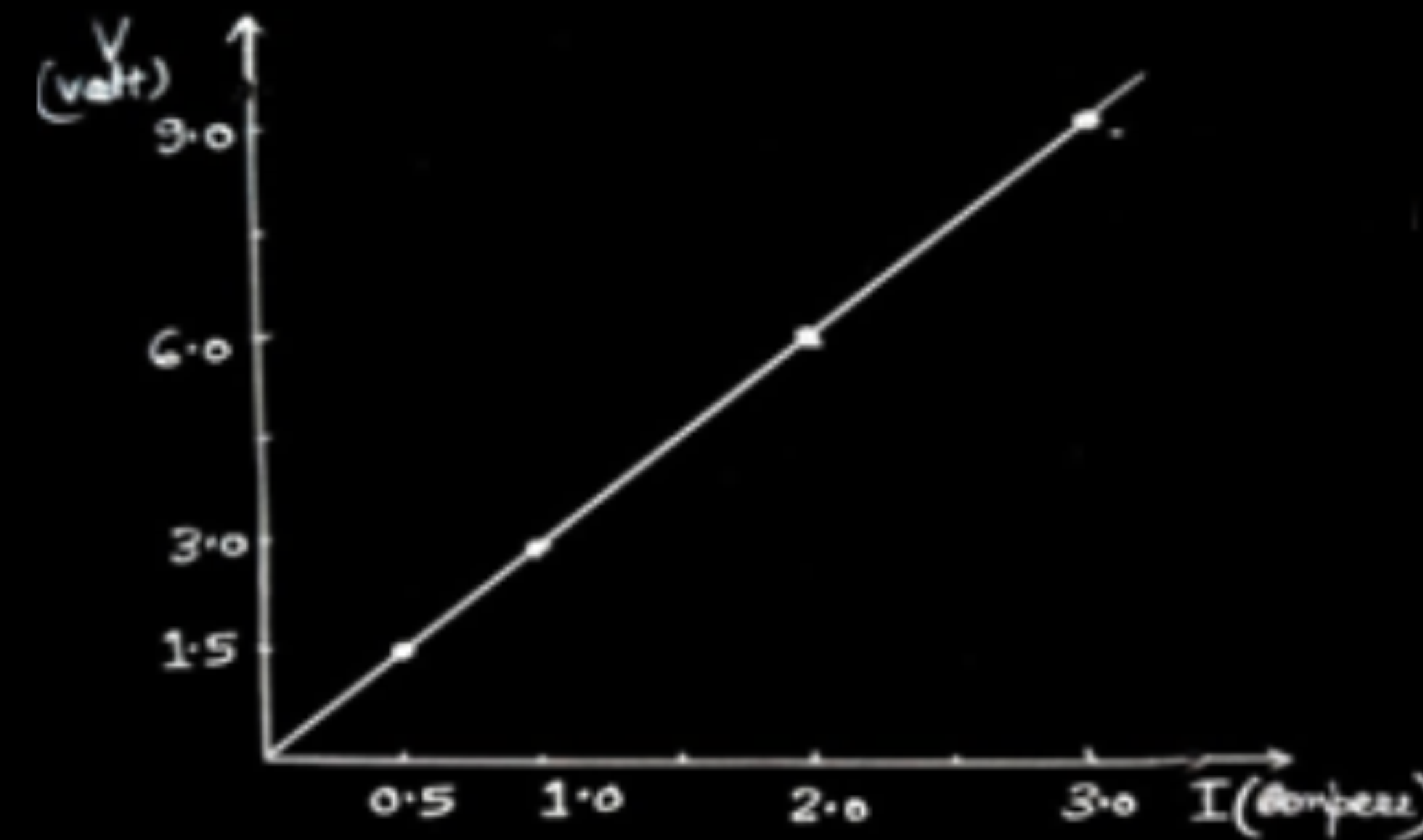
- 10.** (a) The values of current I flowing through a resistor for the corresponding values of potential difference V across it are given below :

V (volts)	1.5	3.0	6.0	9.0
I (amperes)	0.5	1.0	2.0	3.0

CBSE 2022 C

- (i) Plot a graph between V and I .
- (ii) Why should this graph pass through the origin ?
- (iii) Name and state the law which is represented by the graph. 3

(a) (i)



(ii) According to Ohm's law, there should be no current in conductor if potential difference is zero. That is why, this graph be straight line passing through the origin.

(iii) Ohm's Law

The current flowing through a given conductor is directly proportional to the potential difference applied across the ends of conductor provided that temperature of conductor remains constant. / $V \propto I$

1

1

$\frac{1}{2}$

$\frac{1}{2}$

11. (a) (i) In the following figure, three cylindrical conductors A, B and C are shown along with their lengths and areas of cross-section.



If these three conductors are made of the same material and R_A , R_B and R_C be their respective resistances, then find (I) R_A / R_B , and (II) R_A / R_C .

- (ii) If the conductor A is made of copper and the conductor C is made of constantan (alloy of copper and nickel), then which one of the two will have more electrical resistance and why ?

3

(a) (i) (I) $\frac{R_A}{R_B} = \frac{\rho \cdot \left(\frac{L}{A}\right)}{\rho \cdot \left(\frac{L}{4A}\right)}$ $= 4$ (II) $\frac{R_A}{R_C} = \frac{\rho \cdot \left(\frac{L}{A}\right)}{\rho \cdot \left(\frac{L}{2}\right) / \left(\frac{A}{2}\right)}$ $= 1$ (ii) • C • The resistivity of alloys is generally higher than that of the pure metals.	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
---	---

8. (a) Write the relationship between electrical resistance and electrical resistivity for a metallic conductor of cylindrical shape. Hence derive the SI unit of electrical resistivity.

3

(b) Find the resistivity of the material of a metallic conductor of length 2m and area of cross-section $1.4 \times 10^{-6} \text{m}^2$. The resistance of the conductor is 0.04 ohm.

CBSE 2022

8. (a) Write the relationship between electrical resistance and electrical resistivity for a metallic conductor of cylindrical shape. Hence derive the SI unit of electrical resistivity.

3

(b) Find the resistivity of the material of a metallic conductor of length 2m and area of cross-section $1.4 \times 10^{-6} \text{m}^2$. The resistance of the conductor is 0.04 ohm.

CBSE 2022

$$(a) \rho = R \frac{A}{l}$$

$$\begin{aligned} \text{SI unit of } \rho &= \text{ohm} \times \frac{\text{m}^2}{\text{m}} \\ &= \text{ohm} \times \text{metre} / \Omega \text{ m} \end{aligned}$$

(b)

$$\begin{aligned} \rho &= R \frac{A}{l} \\ &= \frac{0.04 \Omega \times 1.4 \times 10^{-6} \text{m}^2}{2 \text{m}} \\ &= 2.8 \times 10^{-8} \Omega \text{m} \end{aligned}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

9. You are provided with a resistor, a key, an ammeter, a voltmeter, four cells of 1.5 V each and few connecting wires. Using these circuit components, draw a labelled circuit diagram to show the setup to study the Ohm's law.

State the relationship between potential difference (V) across the resistor and the current (I) flowing through it. Also draw V-I graph, taking V on the X-axis.

CBSE 2022

Electric circuit

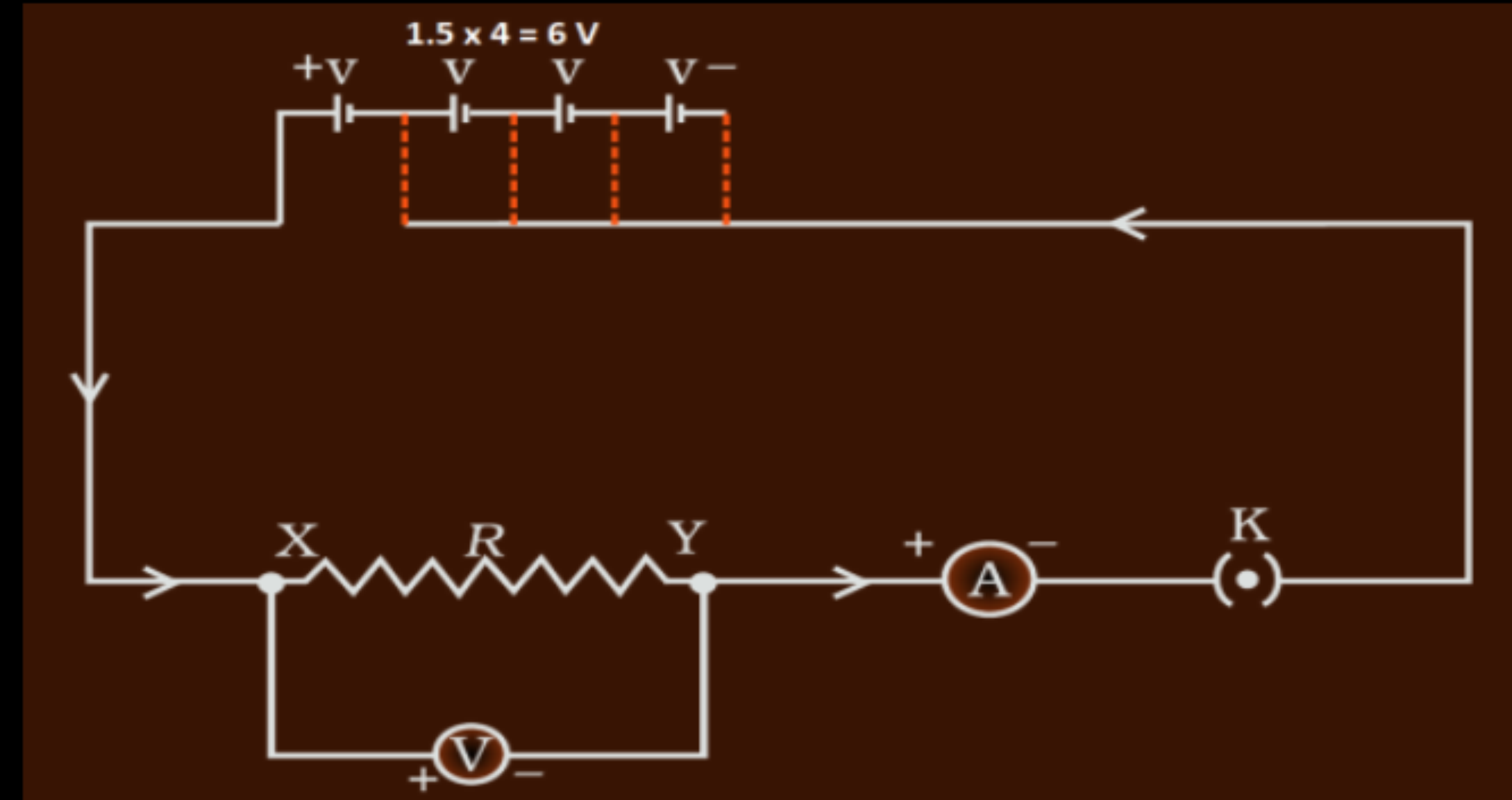
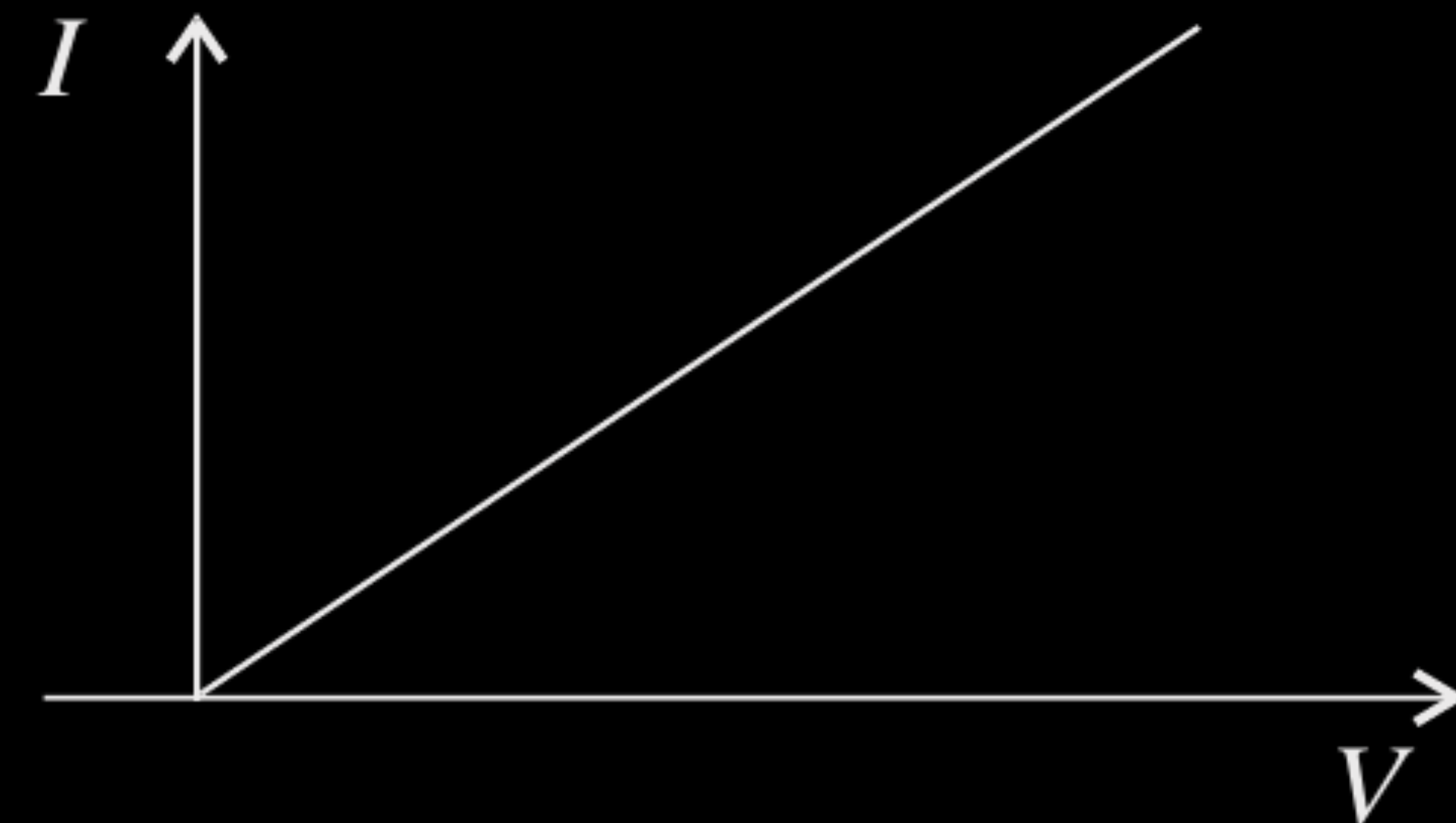


Diagram making

Any two labelling

Potential difference (V) directly proportional to the current (I).



1
 $\frac{1}{2}$, $\frac{1}{2}$
 $\frac{1}{2}$

$\frac{1}{2}$

13. (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. 2+1
- (b) The resistance of a wire of 0.01 cm radius is $10\ \Omega$. If the resistivity of the wire is $50 \times 10^{-8}\ \Omega\text{ m}$, find the length of this wire. CBSE 2022

13. (a) List the factors on which the resistance of a uniform cylindrical conductor of a given material depends. 2+1
- (b) The resistance of a wire of 0.01 cm radius is 10 Ω . If the resistivity of the wire is $50 \times 10^{-8} \Omega \text{ m}$, find the length of this wire. CBSE 2022

(a) (i) Length of the conductor (l)	1
(ii) Area of cross-section of the conductor (A)	1
(b) Radius of wire, $r = 0.01 \text{ cm} = 0.01 \times 10^{-2} \text{ m}$	
Resistance, $R = 10 \Omega$	
Resistivity, $\rho = 50 \times 10^{-8} \Omega \text{ m}$	
$R = \rho \frac{l}{A} = \rho \frac{l}{\pi r^2} \Rightarrow l = \frac{R\pi r^2}{\rho}$	$\frac{1}{2}$
$l = \frac{10\Omega \times 22 \times (0.01 \times 10^{-2})^2 \text{ m}}{7 \times 50 \times 10^{-8} \Omega \text{ m}}$ $= \frac{22}{35} \text{ m} = 0.629 \text{ m} = 0.628 \text{ m} = 0.62 \text{ m}$	$\frac{1}{2}$

11. (a) State Ohm's Law. Represent it mathematically.

3

(b) Define 1 ohm.

CBSE 2022

(c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends ?

11. (a) State Ohm's Law. Represent it mathematically.

3

(b) Define 1 ohm.

CBSE 2022

(c) What is the resistance of a conductor through which a current of 0.5 A flows when a potential difference of 2 V is applied across its ends ?

(a) The potential difference(V) across the ends of a given metallic wire or conductor is directly proportional to the current(I) flowing through it, provided the temperature / physical conditions of the conductor remains the same.

$$V \propto I$$

$$V/I = \text{Constant}$$

$$V = I R$$

$\frac{1}{2}$

$\frac{1}{2}$

(b) If the potential difference across the two ends of a conductor is 1 V and the current through it is 1 A, then the resistance R of the conductor is 1 ohm.

1

$$\text{Alternative answer: } 1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ ampere}}$$

(c)

$$\bullet R = \frac{V}{I}$$

$$R = \frac{2V}{0.5 \text{ A}}$$

$\frac{1}{2}$

$\frac{1}{2}$

• 4 ohm

10. (a) Calculate the resistance of a metal wire of length 2 m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$.
(Resistivity of the metal is $2.8 \times 10^{-8} \Omega \text{ m}$)

(b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices ?

CBSE 2022

2+1

10. (a) Calculate the resistance of a metal wire of length 2 m and area of cross-section $1.55 \times 10^{-6} \text{ m}^2$.
(Resistivity of the metal is $2.8 \times 10^{-8} \Omega \text{ m}$)
- (b) Why are alloys preferred over pure metals to make the heating elements of electrical heating devices ?

CBSE 2022

2+1

(a) Given :

Length of the wire = 2 m

Area of cross section = $1.55 \times 10^{-6} \text{ m}^2$

Resistivity, $\rho = 2.8 \times 10^{-8} \Omega \text{ m}$

$$R = \rho \frac{l}{a}$$

$$= \frac{2.8 \times 10^{-8} \text{ ohm metre} \times 2 \text{ metre}}{1.55 \times 10^{-6} (\text{metre})^2} \Omega$$

$$= 3.6 \times 10^{-2} \Omega$$

(deduct half mark if no or incorrect unit is given)

(b) Alloys used for heating element have generally high melting point / high resistivity / Do not get oxidised at high temperature.

1/2

1/2

1

1

6. The values of mA and μA are

1

- (A) 10^{-6} A and 10^{-9} A respectively
- (B) 10^{-3} A and 10^{-6} A respectively
- (C) 10^{-3} A and 10^{-9} A respectively
- (D) 10^{-6} A and 10^{-3} A respectively

CBSE 2020

6. The values of mA and μA are

1

- (A) 10^{-6} A and 10^{-9} A respectively
- (B) 10^{-3} A and 10^{-6} A respectively
- (C) 10^{-3} A and 10^{-9} A respectively
- (D) 10^{-6} A and 10^{-3} A respectively

CBSE 2020

(B) / 10^{-3} A and 10^{-6} A respectively

7. A cylindrical conductor of length ' l ' and uniform area of cross-section ' A ' has resistance ' R '. Another conductor of length $2.5\ l$ and resistance $0.5\ R$ of the same material has area of cross-section

- (A) $5\ A$
- (B) $2.5\ A$
- (C) $0.5\ A$
- (D) $\frac{1}{5}\ A$

CBSE 2020

7. A cylindrical conductor of length ' l ' and uniform area of cross-section ' A ' has resistance ' R '. Another conductor of length $2.5\ l$ and resistance $0.5\ R$ of the same material has area of cross-section

1

- (A) $5\ A$
- (B) $2.5\ A$
- (C) $0.5\ A$
- (D) $\frac{1}{5}\ A$

CBSE 2020

(A) / 5A

6. A cylindrical conductor of length ' l ' and uniform area of cross section ' A ' has resistance ' R '. The area of cross section of another conductor of same material and same resistance but of length ' $2l$ ' is

CBSE 2020

- (a) $\frac{A}{2}$
- (b) $\frac{3A}{2}$
- (c) $2A$
- (d) $3A$

6. A cylindrical conductor of length ' l ' and uniform area of cross section ' A ' has resistance ' R '. The area of cross section of another conductor of same material and same resistance but of length ' $2l$ ' is

CBSE 2020

- (a) $\frac{A}{2}$
- (b) $\frac{3A}{2}$
- (c) $2A$
- (d) $3A$

1

(c) / 2 A

14. **Assertion (A)** : Alloys are commonly used in electrical heating devices like electric iron and heater.

Reason (R) : Resistivity of an alloy is generally higher than that of its constituent metals but the alloys have low melting points than their constituent metals.

CBSE 2020

14. **Assertion (A)** : Alloys are commonly used in electrical heating devices like electric iron and heater.

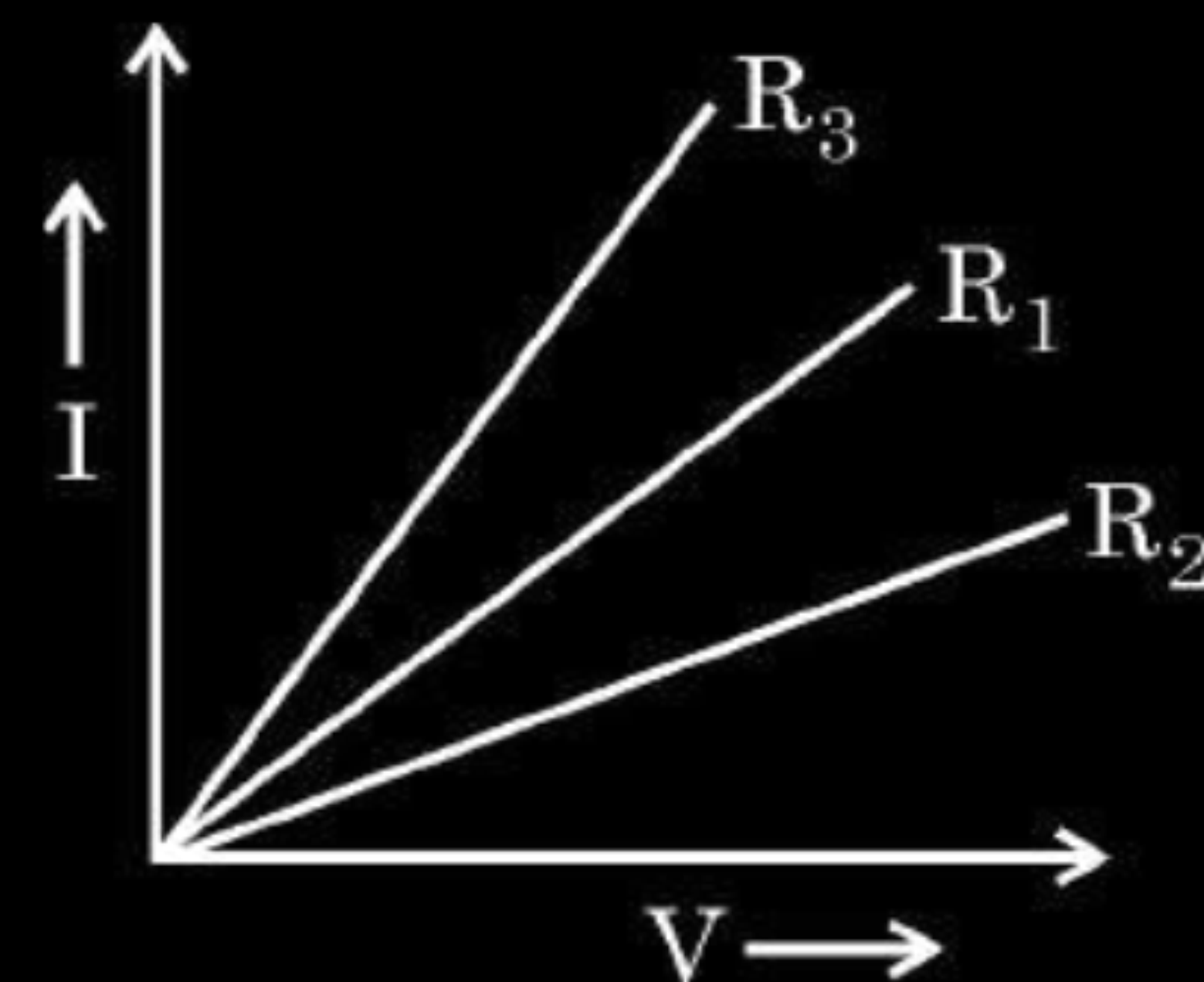
Reason (R) : Resistivity of an alloy is generally higher than that of its constituent metals but the alloys have low melting points than their constituent metals.

1

CBSE 2020

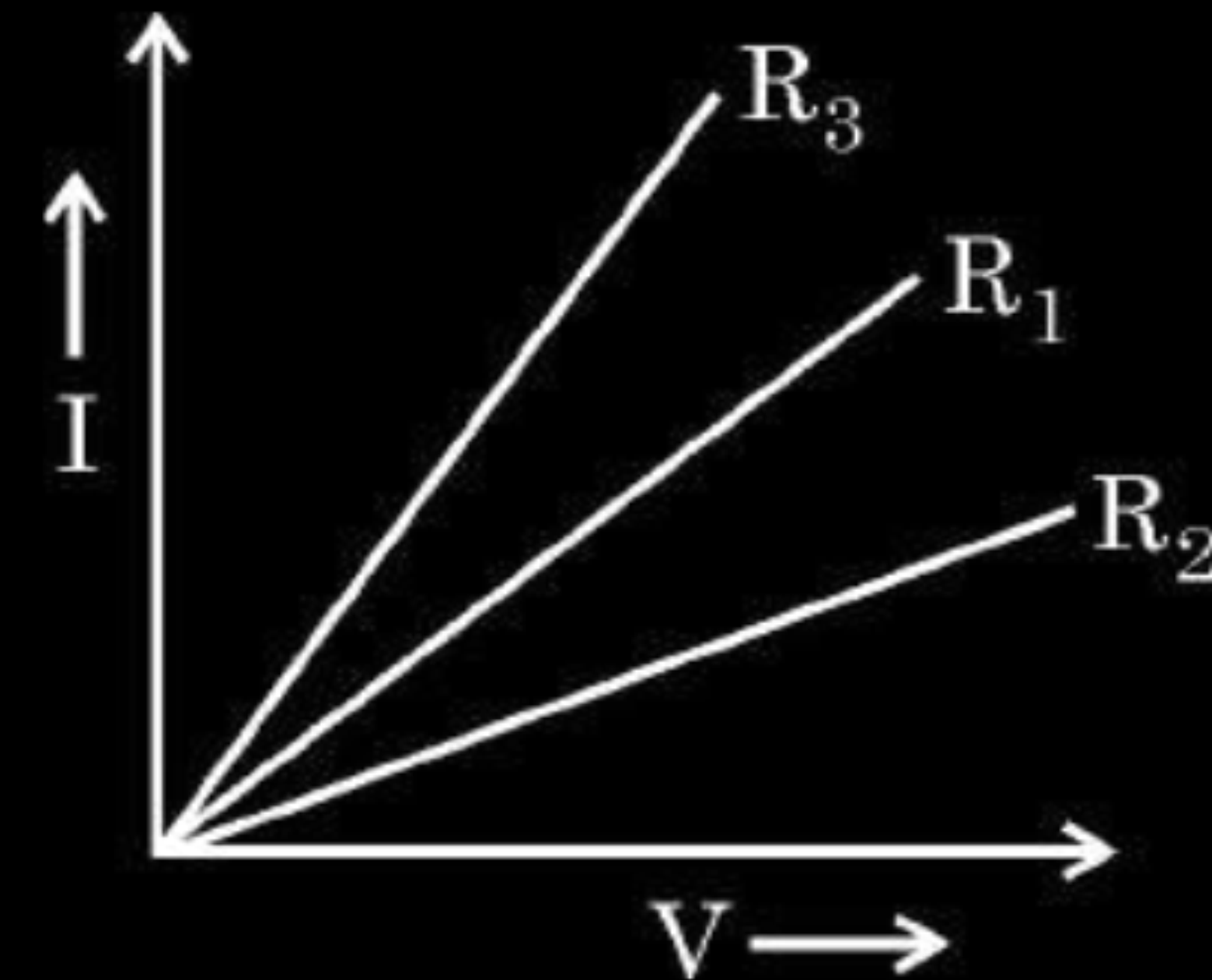
(c) / A is true but R is false.

7. A student plots V-I graphs for three samples of nichrome wire with resistances R_1 , R_2 and R_3 . Choose from the following the statement that holds true for this graph.



- (a) $R_1 = R_2 = R_3$
- (b) $R_1 > R_2 > R_3$
- (c) $R_3 > R_2 > R_1$
- (d) $R_2 > R_1 > R_3$

7. A student plots V-I graphs for three samples of nichrome wire with resistances R_1 , R_2 and R_3 . Choose from the following the statement that holds true for this graph.



- (a) $R_1 = R_2 = R_3$
(b) $R_1 > R_2 > R_3$
(c) $R_3 > R_2 > R_1$
(d) $R_2 > R_1 > R_3$

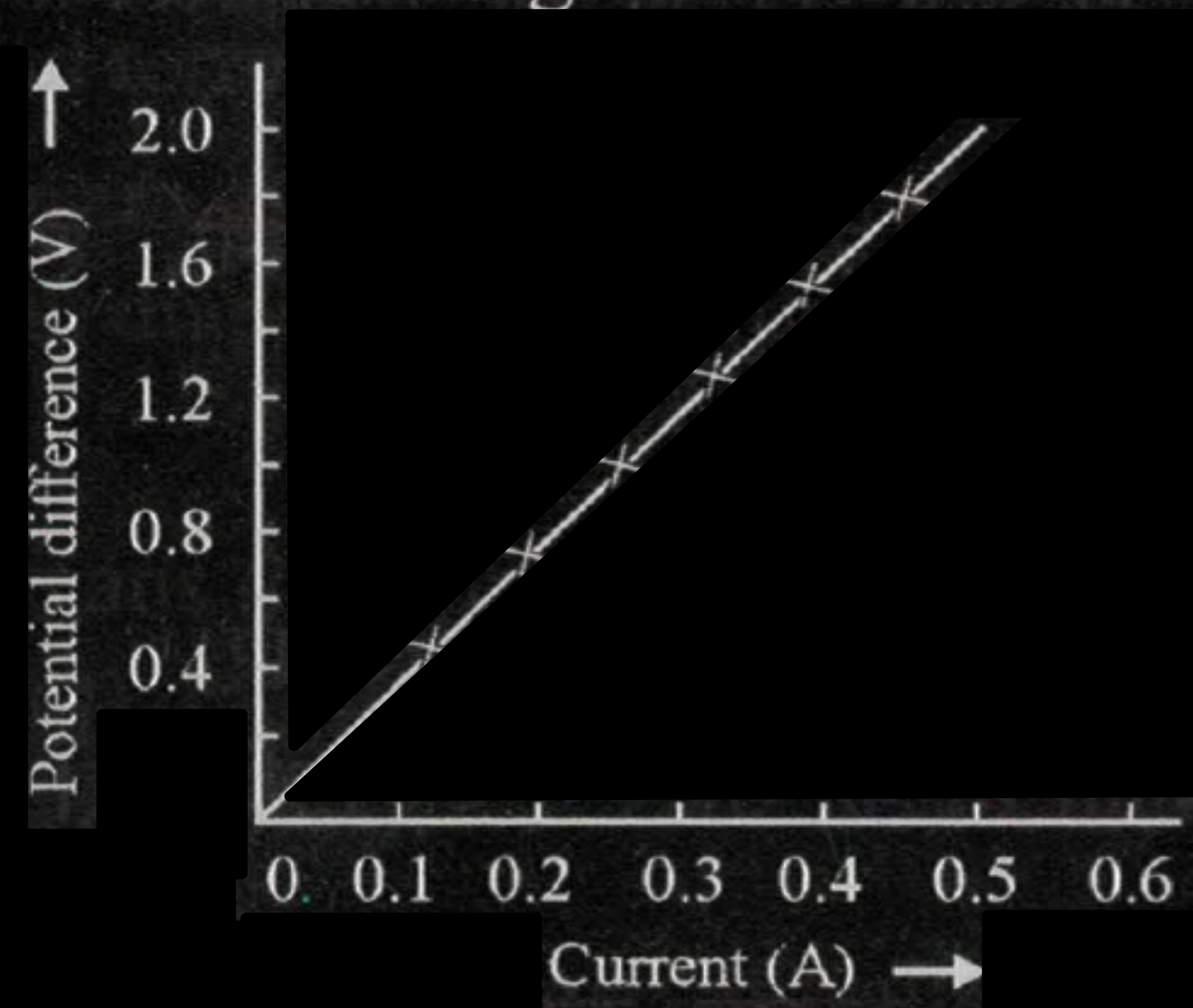
CBSE 2020

1

(d) / $R_2 > R_1 > R_3$

23. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.

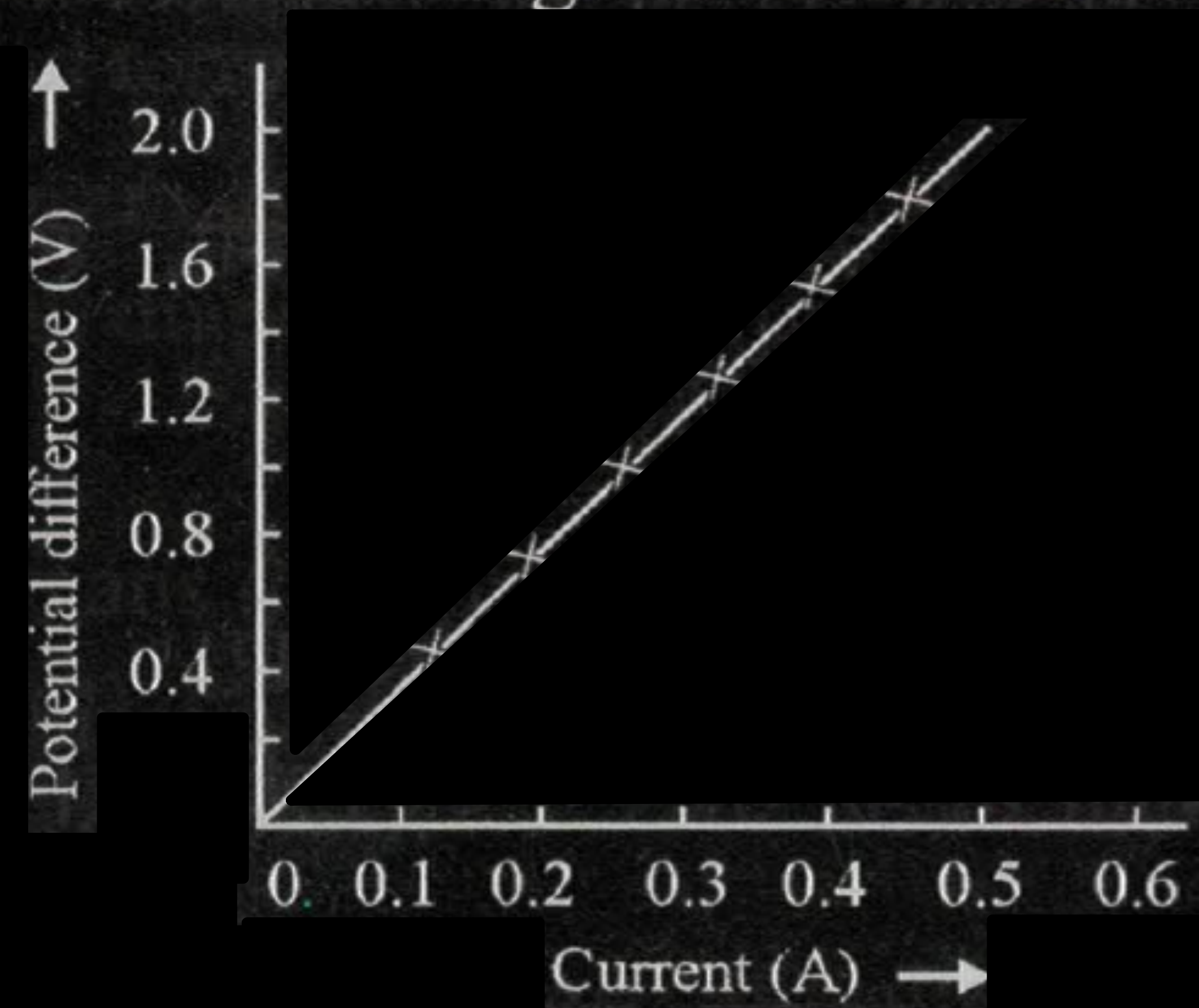
3



CBSE 2020

23. A V-I graph for a nichrome wire is given below. What do you infer from this graph? Draw a labelled circuit diagram to obtain such a graph.

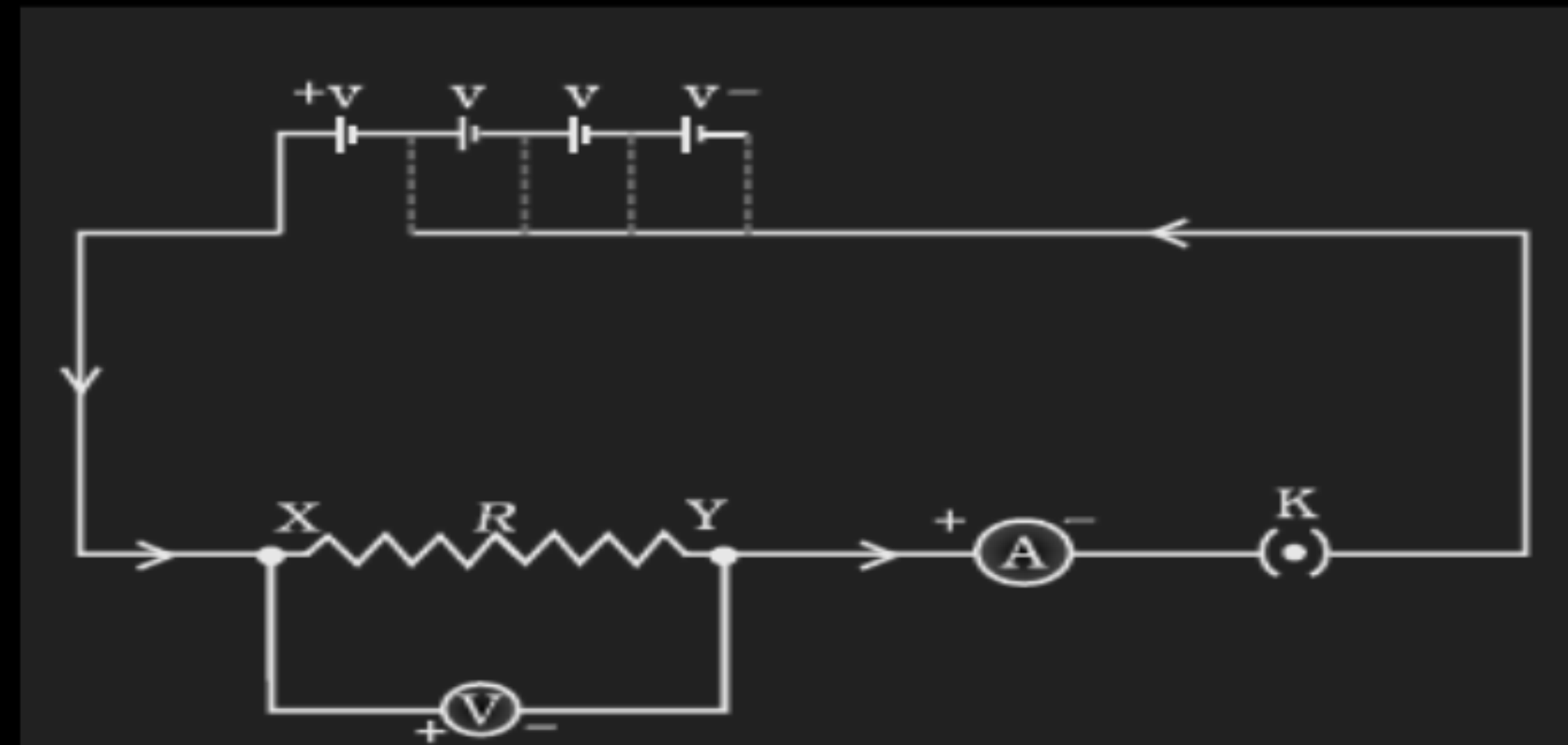
3



CBSE 2020

$V \propto I$ or Potential difference is directly proportional to current

1



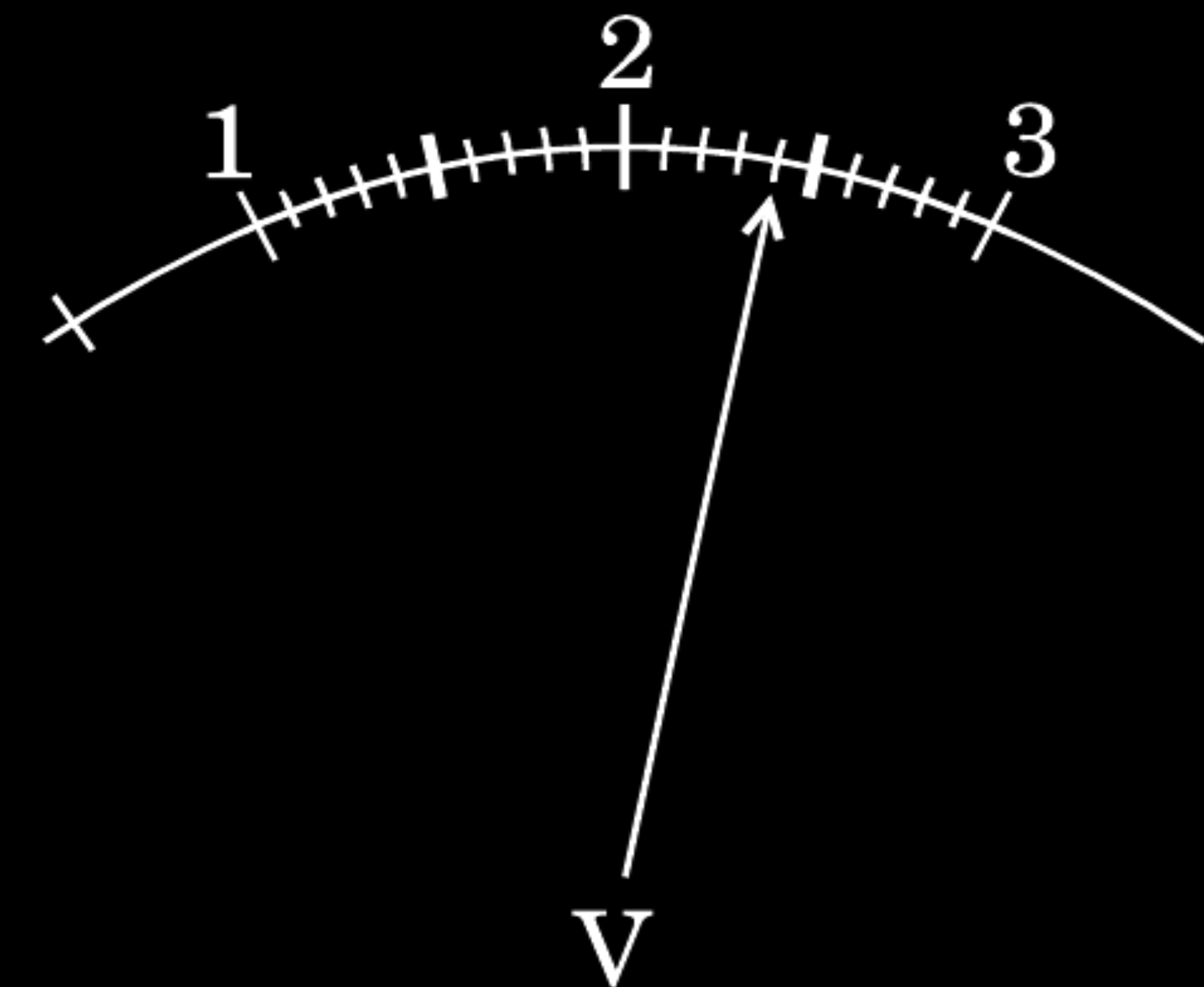
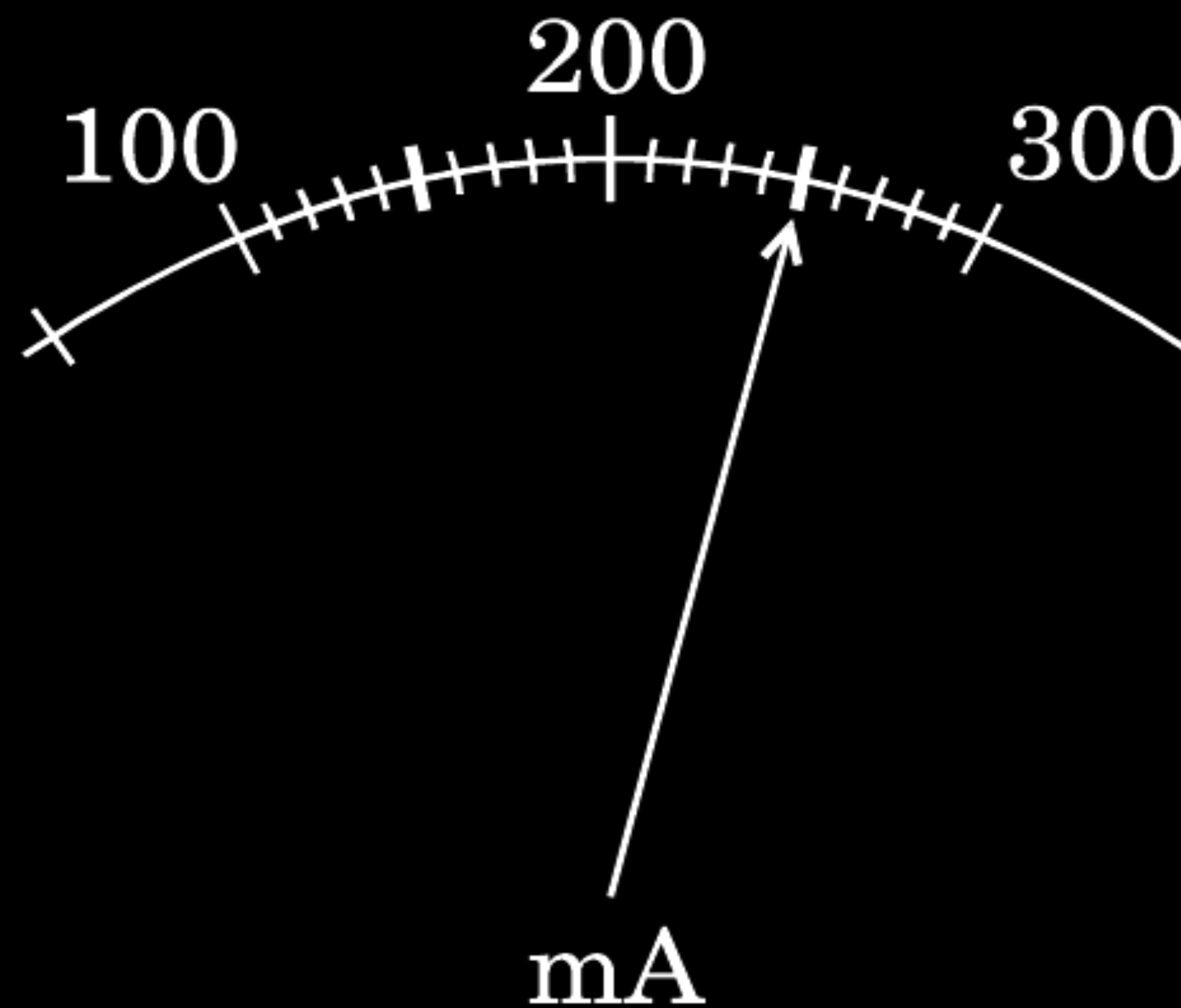
2

Note: If circuit diagram is correct but labelling of ammeter and voltmeter are incorrect, deduct 1 mark.

The current flowing through a resistor connected in a circuit and the potential difference developed across its ends are as shown in the diagram by milliammeter and voltmeter readings respectively :

- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

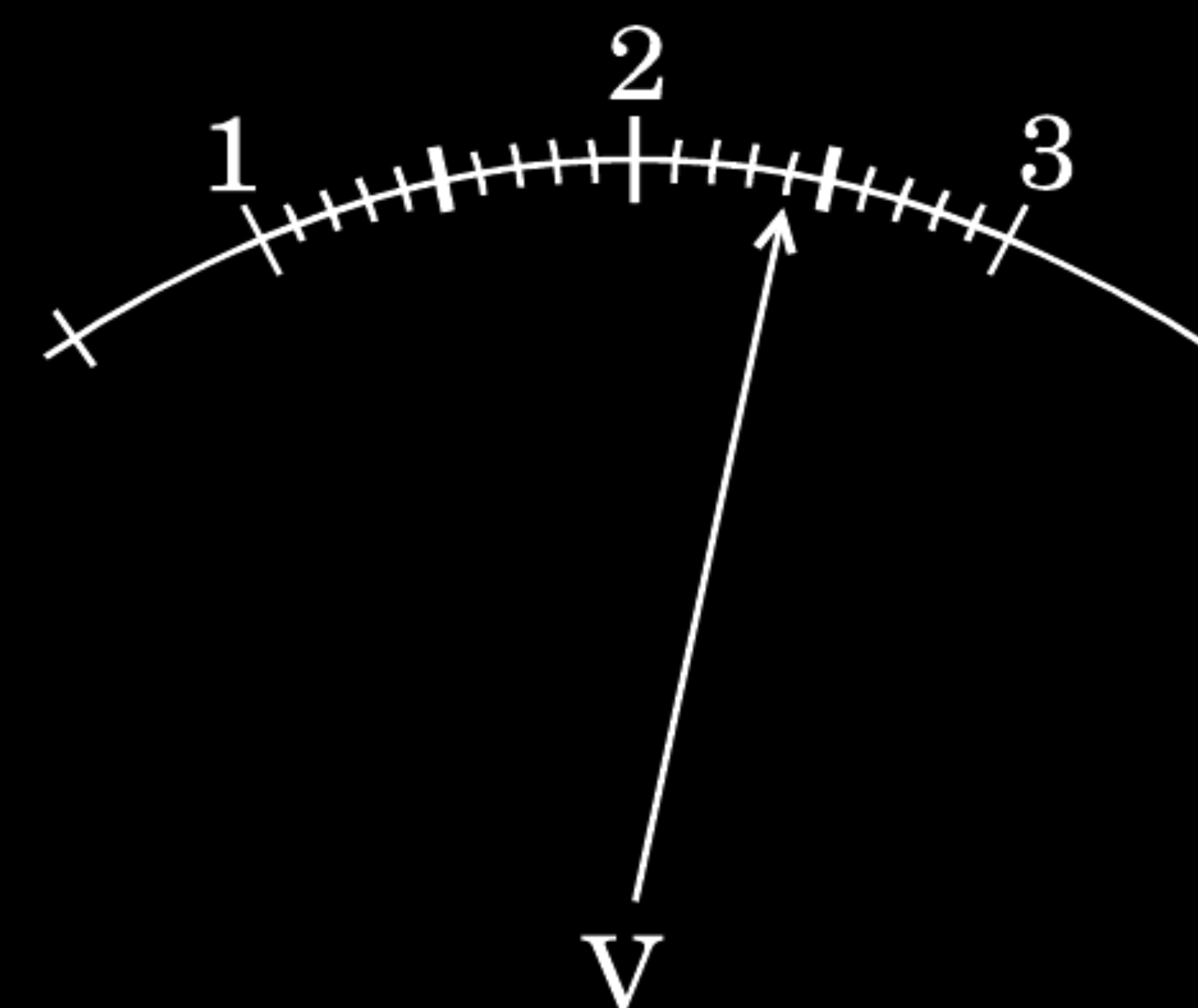
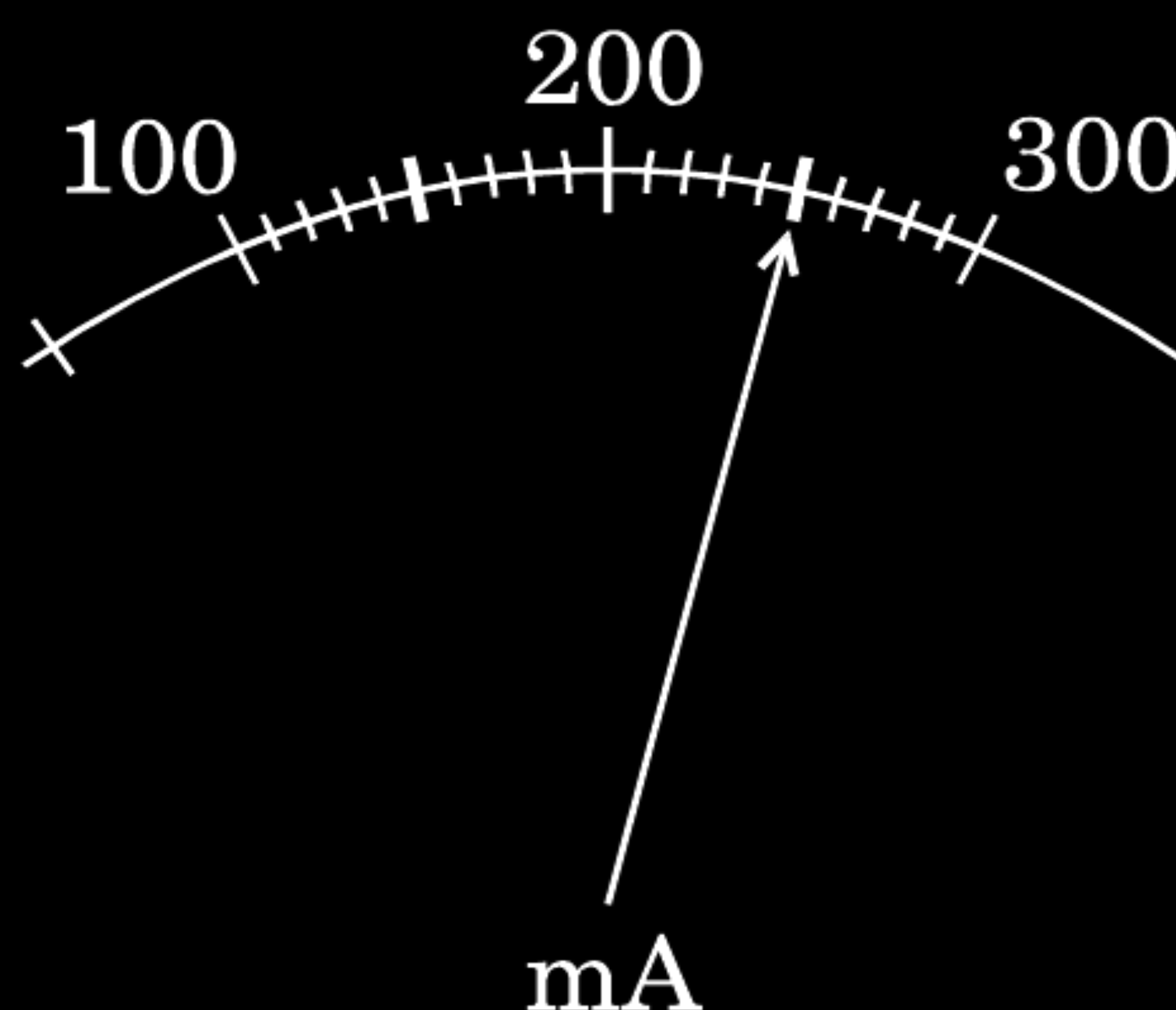
CBSE 2019



The current flowing through a resistor connected in a circuit and the potential difference developed across its ends are as shown in the diagram by milliammeter and voltmeter readings respectively :

- (a) What are the least counts of these meters ?
- (b) What is the resistance of the resistor ?

CBSE 2019



- (a) $\angle$ least count of ammeter = 10 mA
 $\angle$ least count of Voltmeter = 0.1 V

$\frac{1}{2} + \frac{1}{2}$

- (b) $\frac{2.4}{0.25} = 9.6 \text{ ohm (250mA = 0.25A)}$

$\frac{1}{2} + \frac{1}{2}$

Should the resistance of a voltmeter be low or high ? Give reason.

CBSE 2019

Should the resistance of a voltmeter be low or high ? Give reason.

CBSE 2019

High, In parallel connection, less current passes through high resistance.

$\frac{1}{2} + \frac{1}{2}$

Define the term electrical resistivity of a material.

CBSE 2019

Define the term electrical resistivity of a material.

CBSE 2019

Resistance of material / conductor whose area of cross section is 1m^2 and length 1 meter.

1

- (a) Write the relation between resistance and electrical resistivity of the material of a conductor in the shape of a cylinder of length ' l ' and area of cross-section ' A '. Hence derive the S.I. unit of electrical resistivity.
- (b) Resistance of a metal wire of length 5 m is $100\ \Omega$. If the area of cross-section of the wire is $3 \times 10^{-7}\ \text{m}^2$, calculate the resistivity of the metal.

(a)

$$R \propto l$$

$$R \propto \frac{1}{A}$$

$$R \propto \frac{l}{A}$$

$$R = \rho \frac{l}{A}$$

$$\rho = \frac{RA}{l} = \frac{\text{ohm} \times \text{m}^2}{\text{m}}$$

$$= \text{ohm} \times \text{m}$$

$\frac{1}{2} \times 6$

(b)

$$\rho = \frac{RA}{l}$$

$$= \frac{100 \times 3 \times 10^{-7}}{5}$$

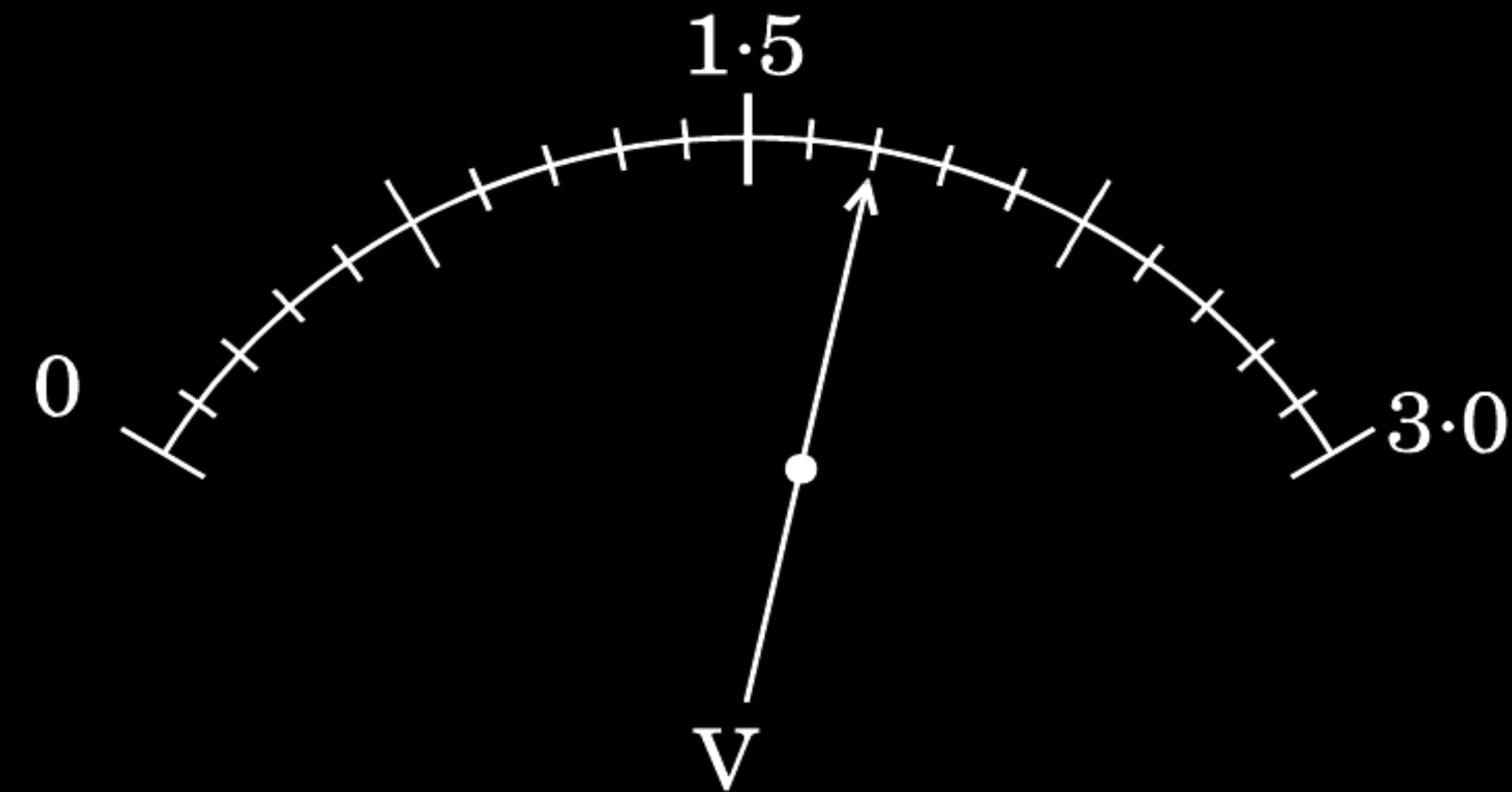
$$= 60 \times 10^{-7} \text{ ohm} \times \text{m}$$

$\frac{1}{2}$

$\frac{1}{2}$

1

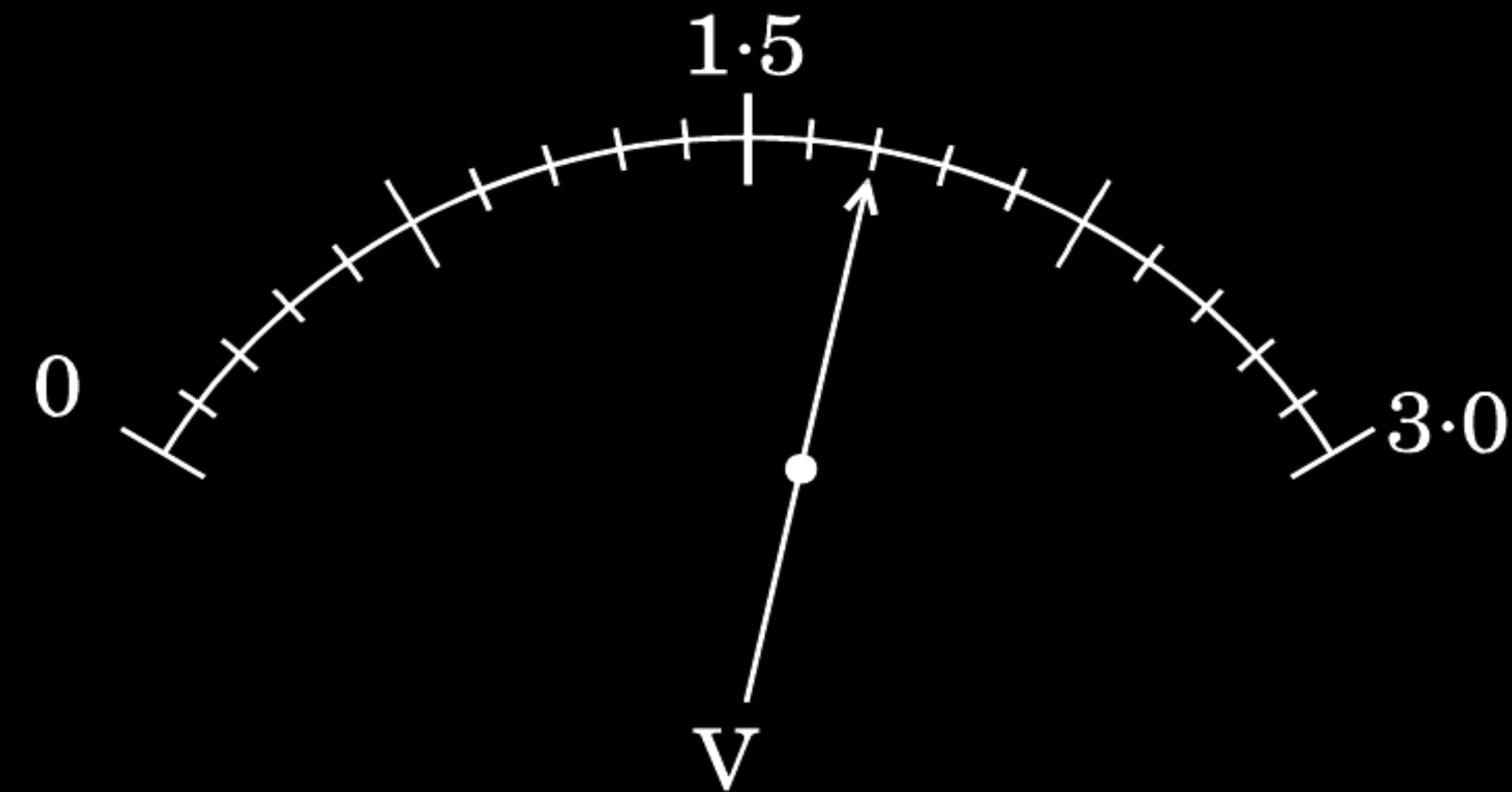
Consider the scale of a voltmeter shown in the diagram and answer the following questions :



- (a) What is the least count of the voltmeter ?
- (b) What is the reading shown by the voltmeter ?
- (c) If this voltmeter is connected across a resistor of $20\ \Omega$, how much current is flowing through the resistor ?

CBSE 2019

Consider the scale of a voltmeter shown in the diagram and answer the following questions :



- (a) What is the least count of the voltmeter ?
- (b) What is the reading shown by the voltmeter ?
- (c) If this voltmeter is connected across a resistor of $20\ \Omega$, how much current is flowing through the resistor ?

CBSE 2019

- a) 0.15V is the least count
- b) The reading shown is 1.8V
- c) $R = 20\Omega$ $V=1.8\text{V}$ $I=\frac{V}{R} = \frac{1.8}{20} = .09\text{amp}$

$\frac{1}{2}$

$\frac{1}{2}$

1

When do we say that the potential difference between two points of a circuit is 1 volt ?

CBSE 2019

When do we say that the potential difference between two points of a circuit is 1 volt ?

CBSE 2019

If 1J of work is done to carry 1 coulomb of charge from one point to another.	1
---	---

Define resistance. Give its S.I. unit.

CBSE 2019

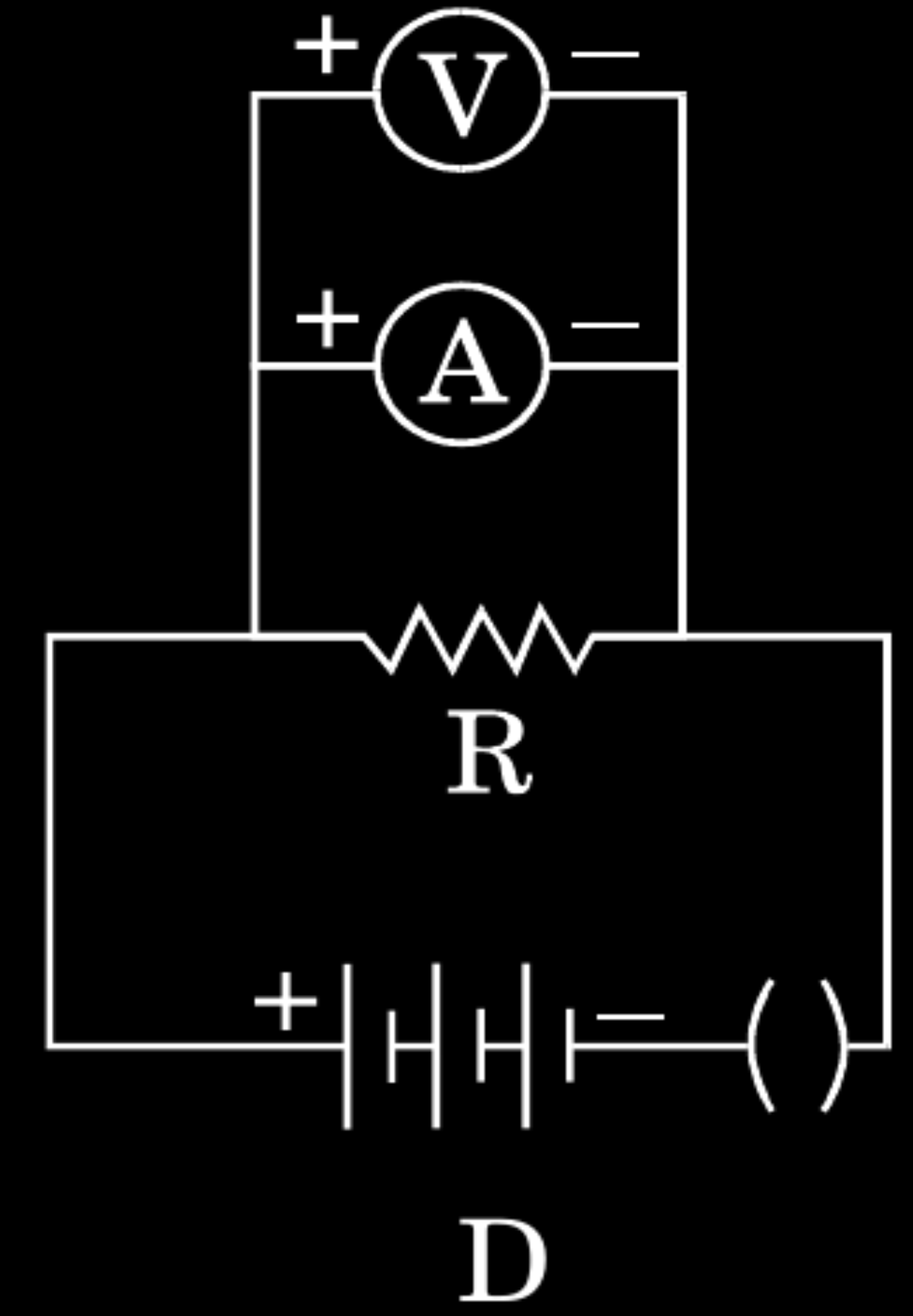
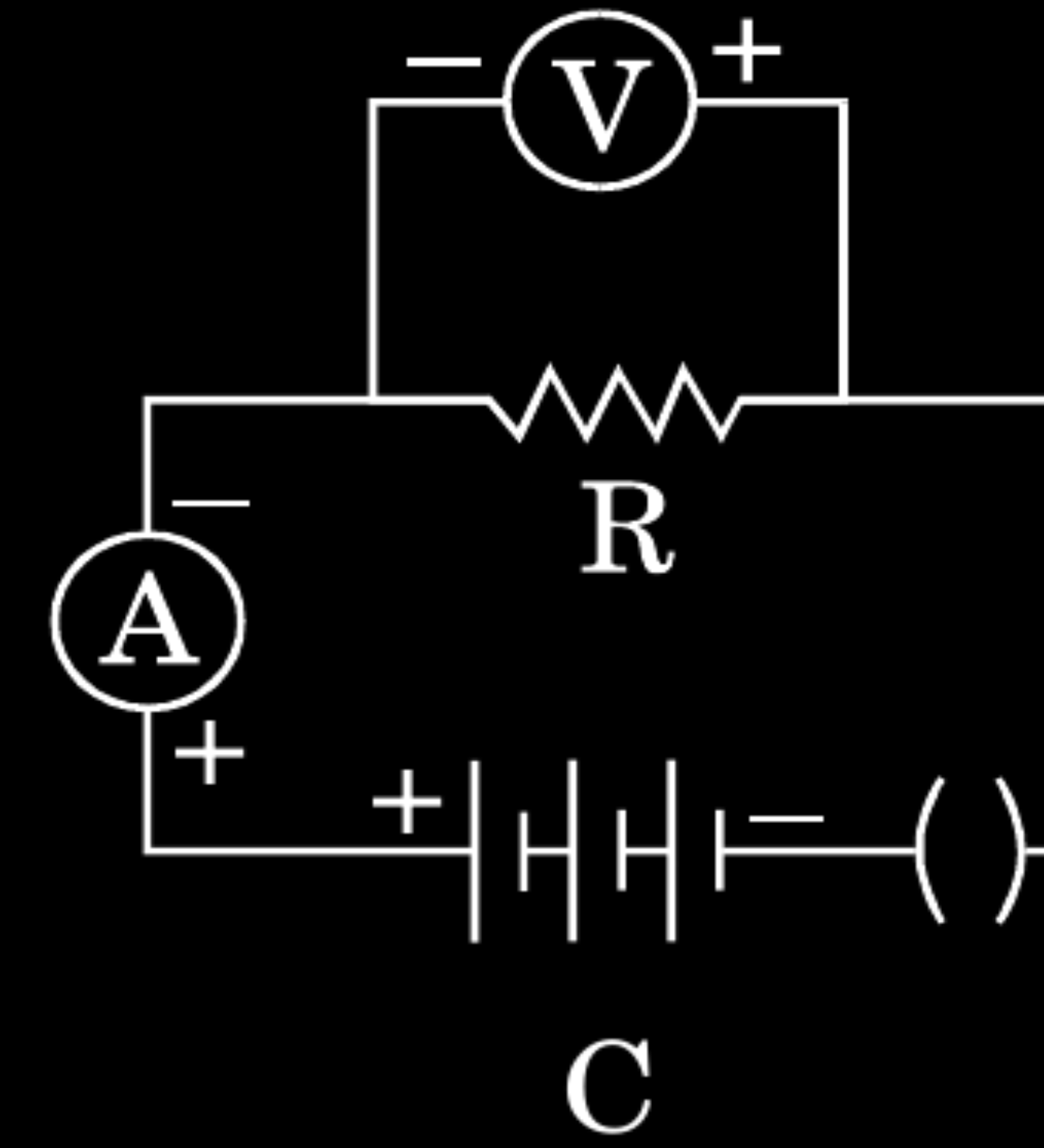
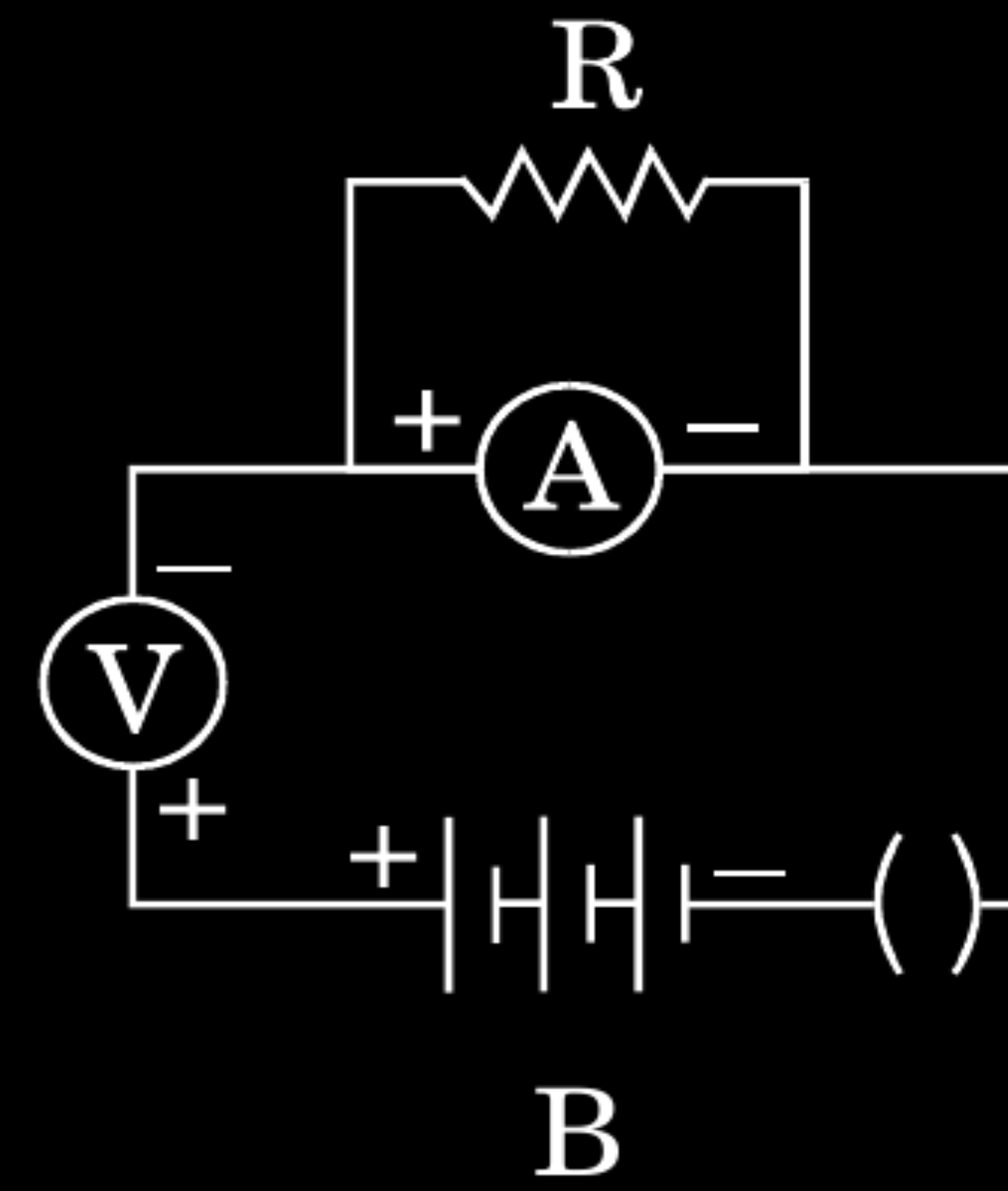
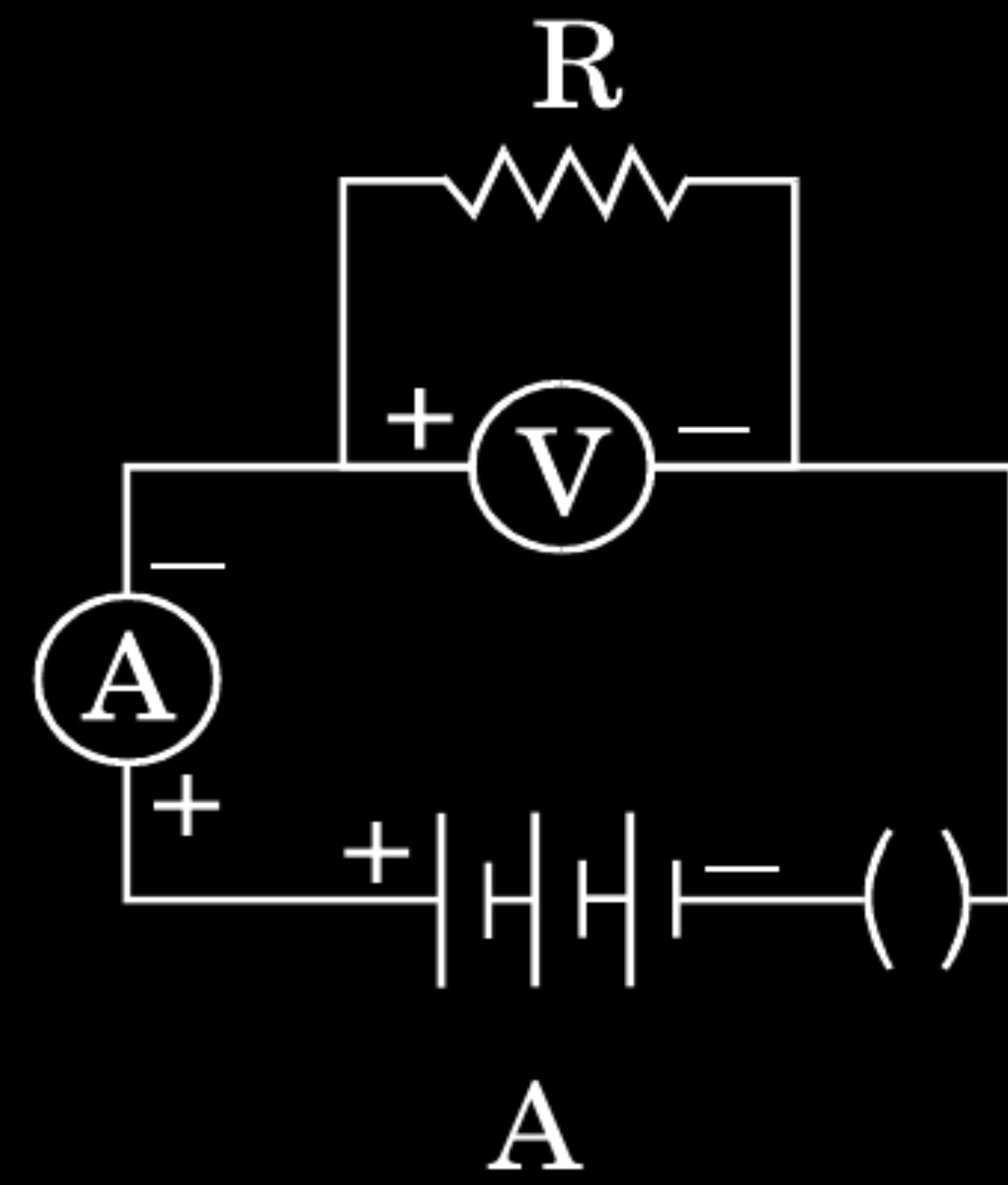
Define resistance. Give its S.I. unit.

CBSE 2019

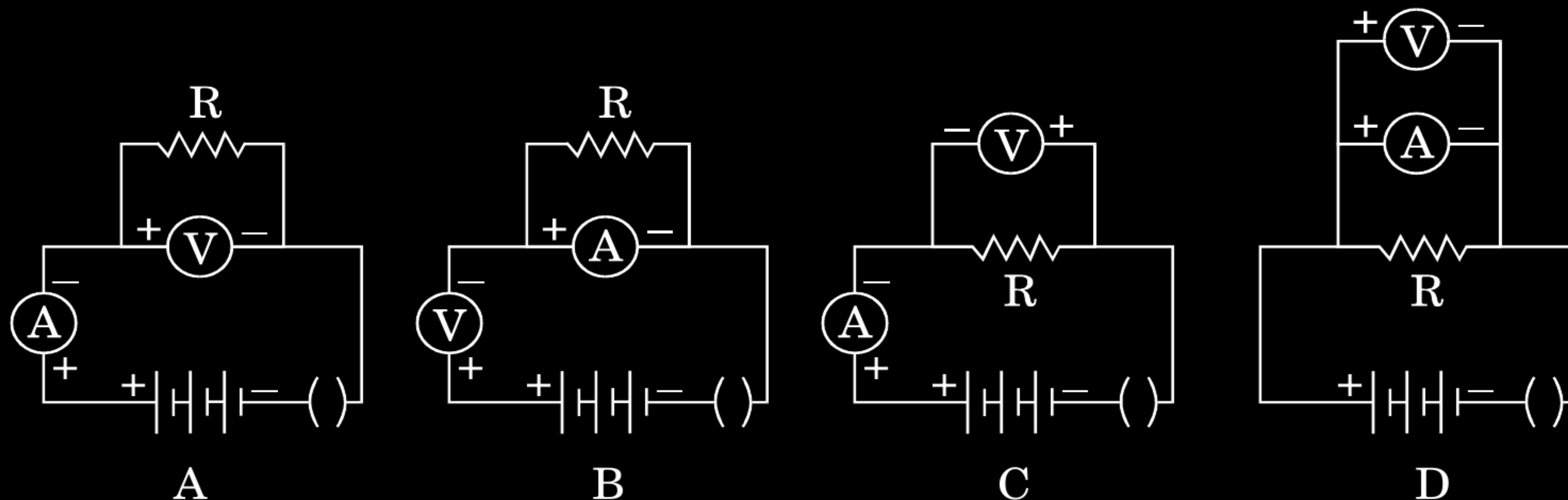
• Resistance is the property of a conductor to resist the flow of charges through it • ohm (Ω)	$\frac{1}{2}$ $\frac{1}{2}$
--	--------------------------------

- (a) Define electric power. An electrical device of resistance R is connected across a source of voltage V and draws a current I . Derive an expression for power in terms of current and resistance.
- (b) Two electric bulbs rated 100 W; 220 V and 60 W; 220 V are connected in parallel to an electric mains of 220 V. Find the current drawn by the bulbs from the mains.

Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



Which one of the following is the correct set-up for studying the dependence of the current on the potential difference across a resistor and why ?



CBSE 2019

- Set up A is correct.
- Ammeter should be connected in series whereas voltmeter should be connected in parallel to the resistor across which potential difference is to be measured.
- Positive of voltmeter and ammeter should be connected to the positive of supply voltage.

$$\frac{1}{2}$$

$$\frac{1}{2} + \frac{1}{2}$$

$$\frac{1}{2}$$

Define current. Give its S.I. unit.

CBSE 2019

Define current. Give its S.I. unit.

CBSE 2019

The rate of flow of charges through a conductor is called electric current. Its S.I. unit is ampere (A)
--

$\frac{1}{2}$

$\frac{1}{2}$

Mention the condition under which a current can flow in a conductor.

CBSE 2019

Mention the condition under which a current can flow in a conductor.

CBSE 2019

Potential difference between ends of conductor	1
--	---

State Ohm's law.

CBSE 2019

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

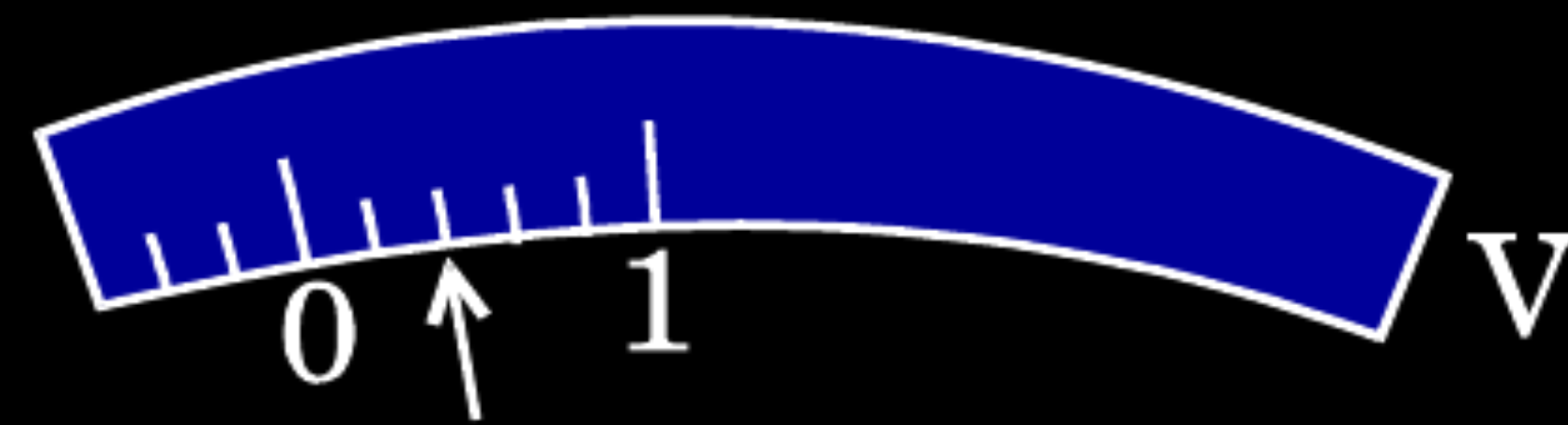
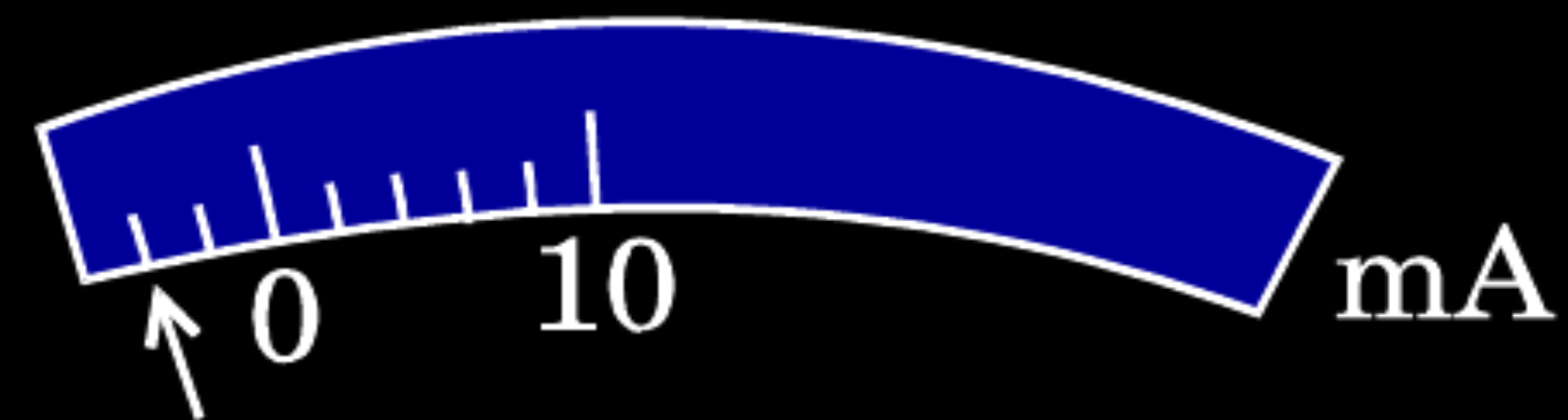


Figure A

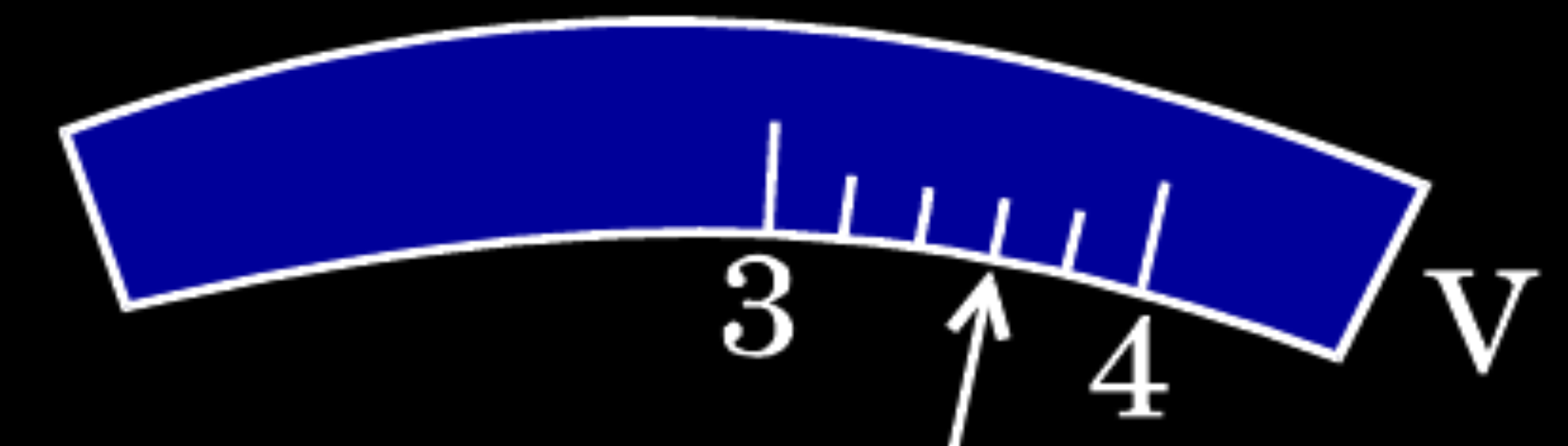
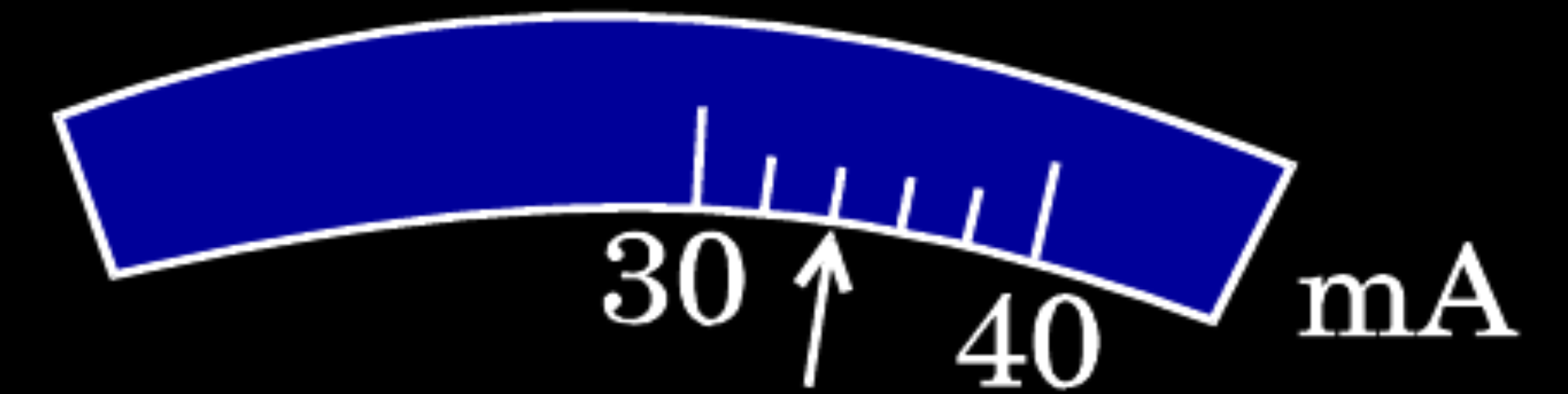


Figure B

The rest position of the needles in a milliammeter and voltmeter, not in use, are as shown in Figure A. When a student uses these instruments in his experiment, the readings of the needles are in the positions shown in Figure B. Determine the correct values of current and voltage the student should use in his calculations.

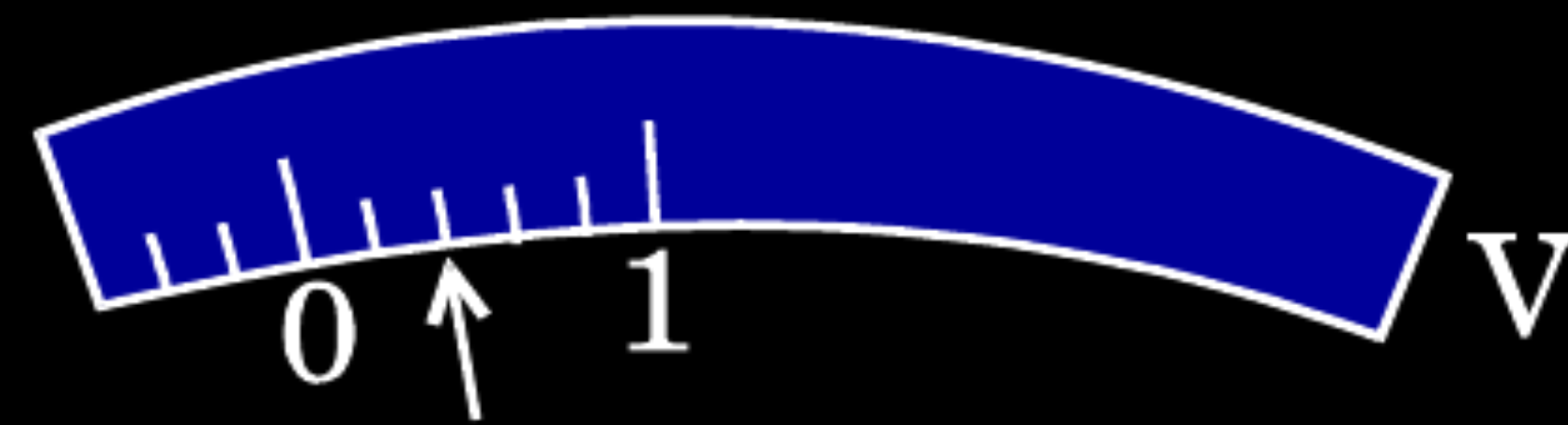
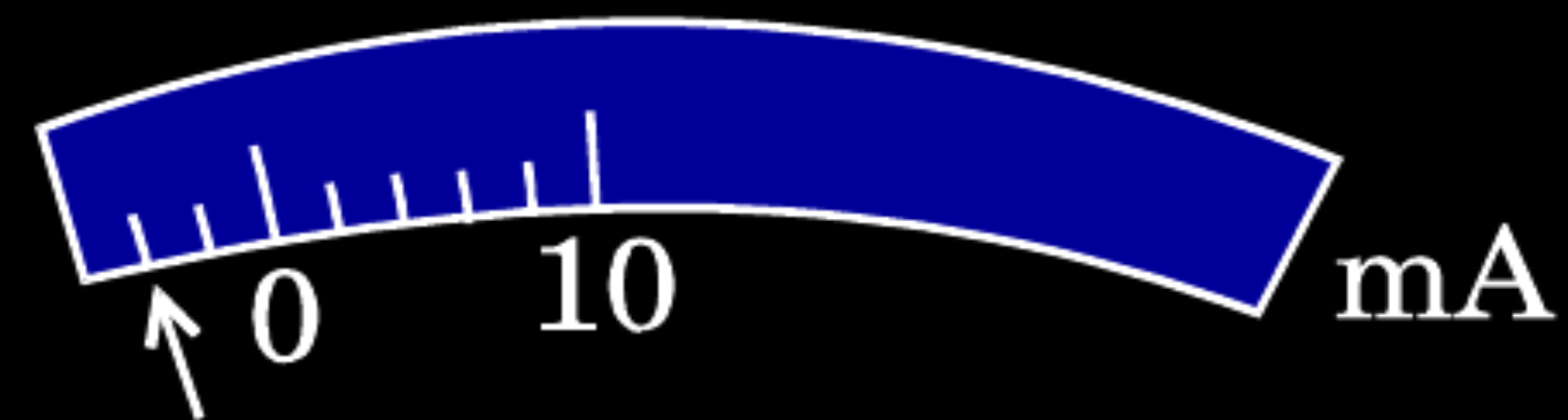


Figure A

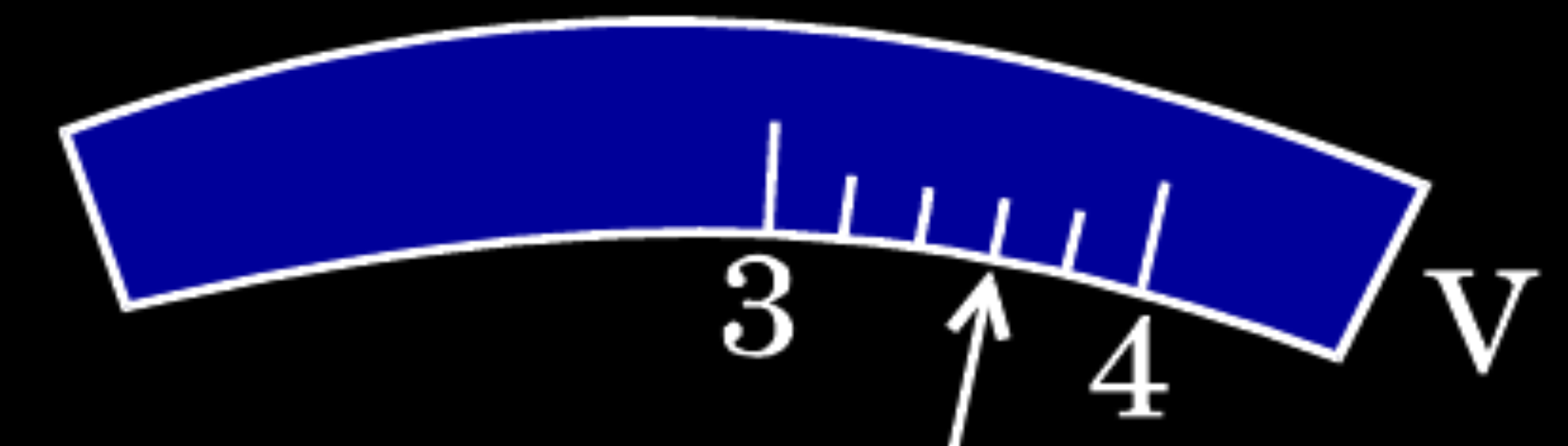
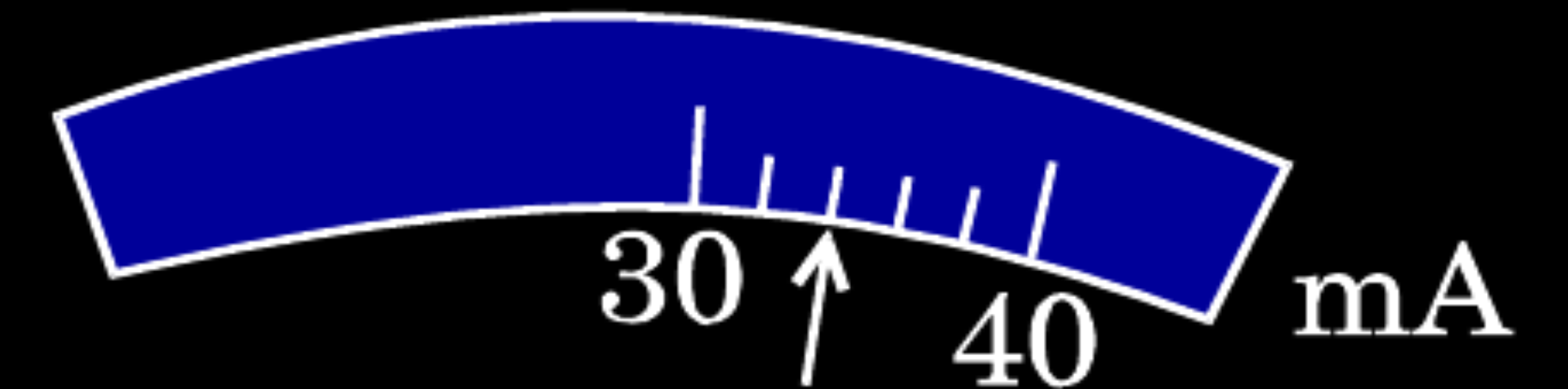


Figure B

CBSE 2019

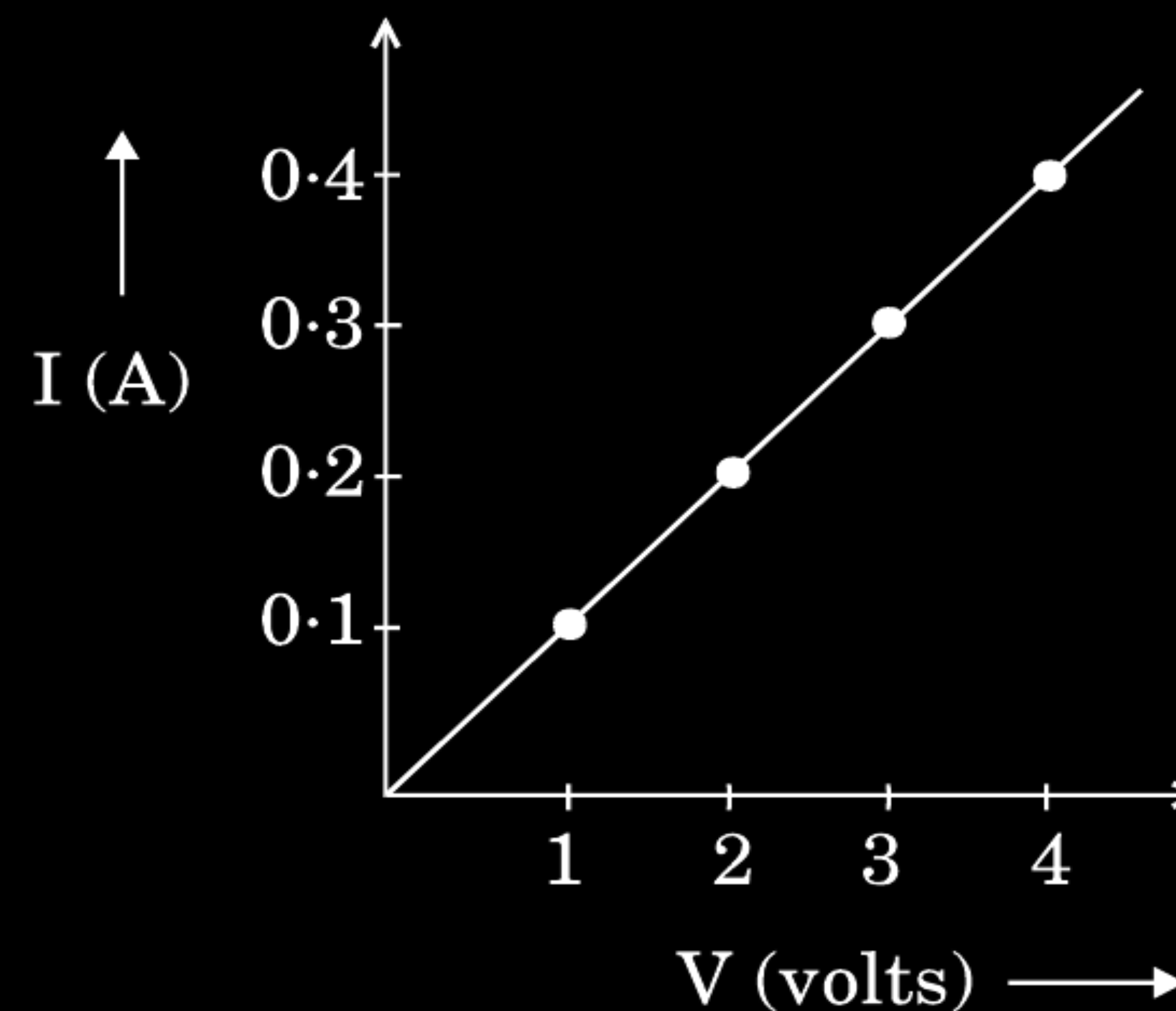
(i) 38mA, 3.2 V

| 1+1

In the experiment to study the dependence of current (I) on the potential difference (V) across a resistor, a student obtained a graph as shown.

- (i) What does the graph depict about the dependence of current on the potential difference ?
- (ii) Find the current that flows through the resistor when the potential difference across it is 2.5 V .

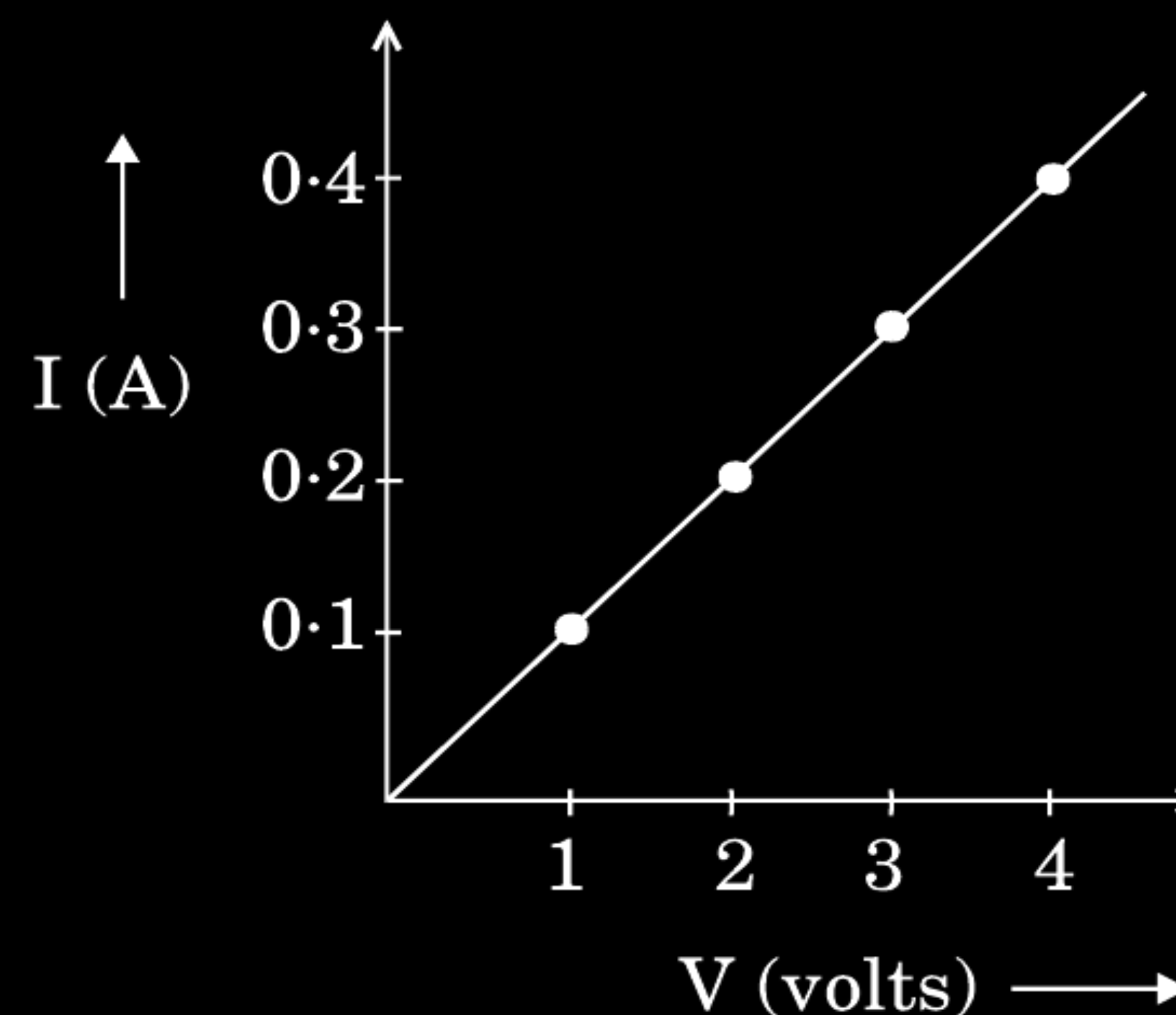
CBSE 2019



In the experiment to study the dependence of current (I) on the potential difference (V) across a resistor, a student obtained a graph as shown.

- (i) What does the graph depict about the dependence of current on the potential difference ?
- (ii) Find the current that flows through the resistor when the potential difference across it is 2.5 V.

CBSE 2019



(i)	$V \propto I$	1
(ii)	at 2.5 V current will be 0.25 A	1

State the S.I. unit of potential difference and name the device used to measure it.

State the S.I. unit of potential difference and name the device used to measure it.

CBSE 2019

- Volt
- Voltmeter

$\frac{1}{2}$

$\frac{1}{2}$

While studying the dependence of potential difference (V) across a resistor on the current (I) passing through it, in order to determine the resistance of the resistor, a student took 5 readings for different values of current and plotted a graph between V and I . He got a straight line graph passing through the origin. What does the straight line signify? Write the method of determining resistance of the resistor using this graph

CBSE 2019

While studying the dependence of potential difference (V) across a resistor on the current (I) passing through it, in order to determine the resistance of the resistor, a student took 5 readings for different values of current and plotted a graph between V and I. He got a straight line graph passing through the origin. What does the straight line signify? Write the method of determining resistance of the resistor using this graph

CBSE 2019

- Potential difference (V) is directly proportional to current (I) or $V \propto I$
- Method: Finding slope of the graph

1

1

What would you suggest to a student if while performing an experiment he finds that the pointer/needle of the ammeter and voltmeter do not coincide with the zero marks on the scales when circuit is open? No extra ammeter/voltmeter is available in the laboratory.

CBSE 2019

What would you suggest to a student if while performing an experiment he finds that the pointer/needle of the ammeter and voltmeter do not coincide with the zero marks on the scales when circuit is open? No extra ammeter/voltmeter is available in the laboratory.

CBSE 2019

- | | |
|---|---|
| • Measure the zero error | 1 |
| • Value of zero error should be adjusted to the observed values | 1 |

Write the function of voltmeter in an electric circuit.

CBSE 2019

Write the function of voltmeter in an electric circuit.

CBSE 2019

To measure potential difference across two points.

Name and define the SI unit of current.

CBSE 2019

Name and define the SI unit of current.

CBSE 2019

- Ampere
- Flow of 1 coulomb of charge per second / 1 ampere = $\frac{1\text{coulomb}}{1\text{second}}$

If the potential difference across the two ends of a conductor is 5 V and the current through it is 0.2 A, then what is the resistance of the conductor ?

CBSE 2019

If the potential difference across the two ends of a conductor is 5 V and the current through it is 0.2 A, then what is the resistance of the conductor ?

CBSE 2019

$$\begin{aligned} R &= \frac{V}{I} \\ &= \frac{5}{0.2} \\ &= 2.5 \, \Omega \end{aligned}$$

1

- (a) List the factors on which the resistance of a conductor in the shape of a wire depends.
- (b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity ? Give reason.
- (c) Why are alloys commonly used in electrical heating devices ? Give reason.

CBSE 2018

- (a) List the factors on which the resistance of a conductor in the shape of a wire depends.
- (b) Why are metals good conductors of electricity whereas glass is a bad conductor of electricity ? Give reason.
- (c) Why are alloys commonly used in electrical heating devices ? Give reason.

CBSE 2018

a. Factors on which resistance of a conductor depends:	
i. Length of conductor [or $R \propto \ell$]	$\frac{1}{2}$
ii. Area of cross-section of the conductor [or $R \propto 1/A$]	$\frac{1}{2}$
b. Metals are good conductor of electricity –as they have low resistivity/ have free electrons	$\frac{1}{2}$
Glass is a bad conductor of electricity – as it has high resistivity/have no free electrons	$\frac{1}{2}$
c. Reason:	
Alloys have high resistivity /high melting point / alloys do not oxidize (Or burn) readily at high temperatures.	
(any one)	1

The values of current (I) flowing through a given resistor of resistance (R), for the corresponding values of potential difference (V) across the resistor are as given below :

V (volts)	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0
I (amperes)	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0

CBSE 2018

Plot a graph between current (I) and potential difference (V) and determine the resistance (R) of the resistor.

The values of current (I) flowing through a given resistor of resistance (R), for the corresponding values of potential difference (V) across the resistor are as given below :

V (volts)	0.5	1.0	1.5	2.0	2.5	3.0	4.0	5.0
I (amperes)	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0

CBSE 2018

Plot a graph between current (I) and potential difference (V) and determine the resistance (R) of the resistor.

